Air Power:
Ground Unit and Ground Target Data

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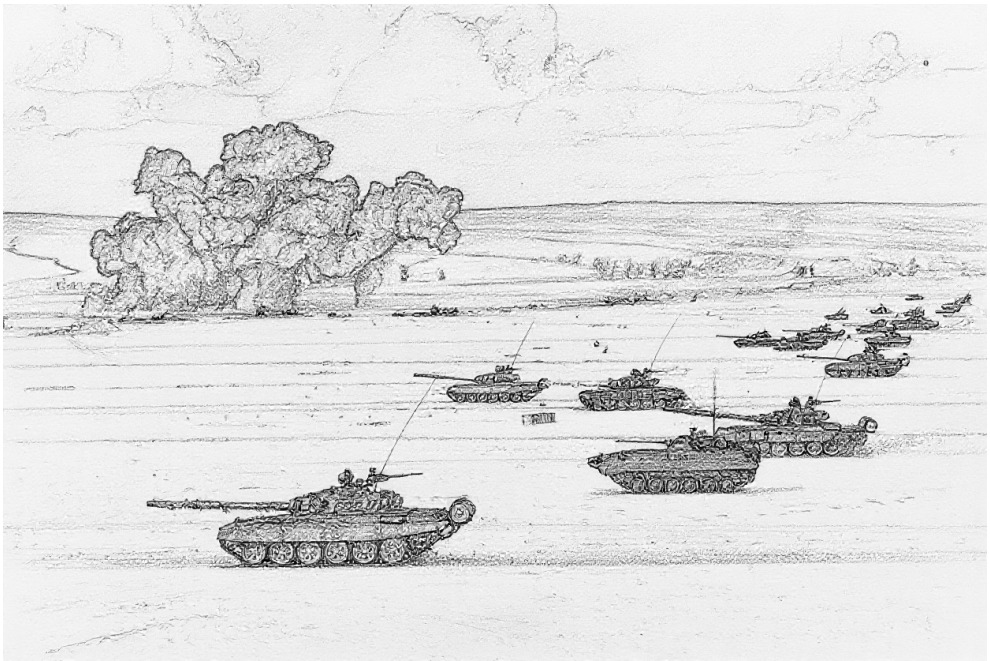


INTRODUCTION

For many, the principal focus of the *Air Power* game is air-to-air combat. That said, no less an authority on air-to-air combat than Robin Olds stated:[1]

> "You're not going to win a war by shooting down a damn MiG, you're going to win a war by bombing the bejesus out of whatever you were sent to bomb."

Here we deal with those "whatevers," the ground units and ground targets that are the targets of air-to-ground attacks, and some of which shoot back.

This document first presents ground-combat and transport units, then air-defense units (AAA, FCR, SAM, EWR, and CCU units), and finally ground targets. An appendix gives some examples of battalion-sized units.

Much of the data here is adapted from *Air Strike*, *Eagles of the Gulf*, and *The Speed of Heat*, from *Air Power Journal*, and from Malcolm Pipe's extensive compendium of AAA and SAM units. However, the data have been carefully revised to improve their accuracy and internal consistency, a wider variety of ground-combat units are covered to give a richer experience, and the descriptions of the units and their weapon systems have been expanded.

Extensive notes accompany the main text, providing provenance of data, explanations of changes, sources of information, and justifications for decisions.

CONTENTS

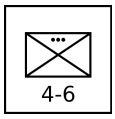
GROUND-COMBAT AND TRANSPORT UNITS

Ground-combat units are formations of infantry, tanks, IFVs, APCs, other AFVs, or artillery whose primary role is combat with other ground units. Transport units provide strategic mobility to infantry and towed artillery and are the life-blood of logistic networks. These units are presented below with their counters. The Ground-Combat and Transport Unit Table summarizes their properties. Some representative examples of vehicles are given in the descriptions, and Vehicle Types Table gives a more extensive list. All ground-combat and transport platoons and batteries are also available as sections and squads.[2]

INFANTRY UNITS

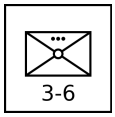
Infantry are soldiers trained to fight on foot and, in some cases, from IFVs. Their primary roles in modern warfare are seizing, clearing, and holding ground, often as part of a combined-arms formation with tanks and artillery, highly mobile offensive and defensive operations in combination with helicopters, and combat in difficult terrain, including urban areas, mountainous zones, and dense jungle or forest.

An important secondary characteristic of infantry is any associated transport. Motorized infantry units are transported by trucks, giving them operational mobility. Mechanized infantry are transported by APCs or IFVs, giving them both tactical and operational mobility. Light infantry units have no or fewer supporting vehicles, although modern light infantry is often airborne.

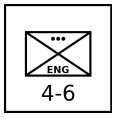


Infantry Platoon: An infantry platoon is armed with small arms, light machine guns, and sometimes with light grenade launchers and portable anti-tank weapons such as bazookas, LAWs, MAWs, and portable ATGMs. It is the primary combat unit in larger infantry formations.

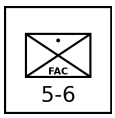
Infantry Weapons Platoon: An infantry weapons platoon is equipped with heavier weapons, such as medium or heavy machine guns, recoilless rifles, portable ATGMs, and automatic grenade launchers. It is often a company- or battalion-level asset, and its role is to provide direct-fire support to the infantry platoons in its formation.[3]



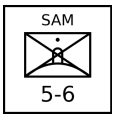
Infantry Mortar Platoon: An infantry mortar platoon is equipped with 60-mm, 81-mm, 82-mm, or similar mortars. It is often battalion-level assets, and its role is to provide indirect fire support and smokescreens to the infantry platoons in its formation. Due to the weight of ammunition required for sustained fire, it will often be transported by a light truck platoon even if the other infantry units in its formation do not have transport.



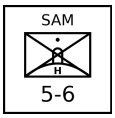
Infantry Engineers Platoon: An infantry engineers platoon is equipped and trained for combat engineering tasks such as mine-laying and mine-clearing. It is often battalion-level asset.



Infantry FAQ Squad: An infantry FAC squad has a forward air controller with other soldiers providing communication and protection. An infantry FAC squad will often form a composite platoon with an infantry or infantry weapons platoon. If operating independently, it will often be transported by a light truck squad.



Infantry SAM Squad: Infantry SAM squads are equipped with shoulder-launched SAMs, with one launcher per squad. Their SAM capabilities and VP values are given elsewhere. An infantry SAM squad will often form a composite platoon with an infantry or infantry weapons platoon. If operating independently, it will often be transported by a light truck squad.



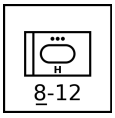
Infantry Heavy SAM Squad: Infantry heavy SAM squads are equipped with pedestal-mounted SAMs, with one launcher per squad. Their SAM capabilities and VP values are given in a subsequent section. An infantry heavy SAM squad will often form a composite platoon with an infantry or infantry weapons platoon. If operating independently, it will often be transported by a light truck squad.

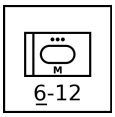
Infantry HQ Platoon: An infantry HQ platoon is the HQ unit of an infantry battalion or regiment. It consists of the formation commander, their staff, and their protection detail.[4]

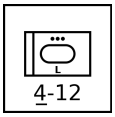
ARMORED VEHICLE UNITS

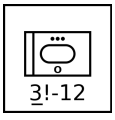
Armored ground-combat vehicle units combine armored protection for their crew and weapons, mobility, and either a firepower or transport capability.

The primary factors that determine the type of an armored-vehicle unit are its weapons and protection level. The means of traction is a secondary consideration. Armored cars and armored half-tracks are considered to be light tanks, light IFVs, or light or open APCs, light anti-tank missile vehicles, light reconnaissance vehicles according to their role and armament.

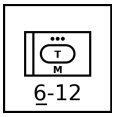


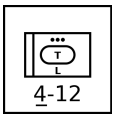




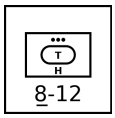


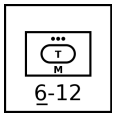
Tank Platoon: A tank platoon has armored vehicles principally armed with a direct-fire gun. The gun is most commonly mounted in a turret, but AFVs with casemate guns such as the SU-100, Kanonen Jpz, and Stridsvagn 103 are also considered to be tanks. The role of a tank is the support of infantry and the destruction of other tanks and armored vehicles. A tank platoon may have heavy, medium, light, or open armor. *Heavy examples:* M1 Abrams, Chieftain, Challenger 1/2, Leopard 2, T-64, and T-72/80. *Medium examples:* M60, Centurion, Leopard 1, T-34, T-54/55, and T-62. *Light examples:* M114A2, M551 Sheridan, Fox, Scimitar, Scorpion, and PT-76. *Open example:* ASU-57.[5]

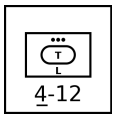


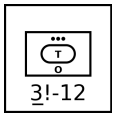


IFV Platoon: An infantry fighting vehicle (IFV) platoon is equipped with armored vehicles that transport infantry and provide them with fire support, being armed with a 20 mm or larger cannon and sometimes an anti-tank missile. An IFV platoon may have medium or light armor. *Medium examples:* M2A2 Bradley and Marder. *Light examples:* M2/M2A1 Bradley, Warrior, and BMP-1/2.[6]

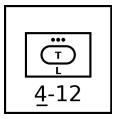


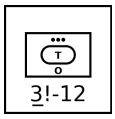




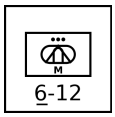


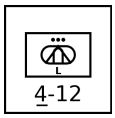
APC Platoon: An APC platoon is equipped with armored vehicles that transport infantry, but unlike IFVs are not equipped to provide fire support and at most are armed with a heavy machine gun. An APC platoon may have heavy, medium, light, or open armor. Heavy and medium APCs are typically converted tanks. *Heavy example:* Nakpadon. *Medium example:* Nagmachon. *Light examples:* M113, FV432, Spartan, and BTR-50PK/60PB/70. *Open examples:* M3 half-track and BTR-50P/60P.[7]



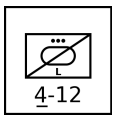


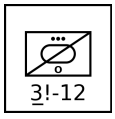
Armored Truck Platoon: An armored truck platoon has armored wheeled vehicles that can transport medium or heavy loads of infantry, towed weapons, and supplies, including fuel and ammunition. Unlike a truck platoon, it has armored protection, but unlike an APC platoon, it has more limited mobility. An armored truck platoon may have light or open armor. *Light example:* BTR-152K. *Open examples:* BTR-40, other BTR-152s, and Saxon.[8]



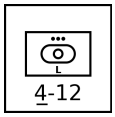


Armored Anti-Tank Missile Platoon: An armored anti-tank missile platoon has armored vehicles equipped with anti-tank guided missiles. Its mission is the destruction of other armored vehicles. Conventional gun-armed tank destroyers such as the Kanonen Jpz are considered to be tanks. An armored anti-tank missile platoon is often a battalion- or regiment-level asset. It may have medium or light armor. *Medium examples:* Rakaten Jpz and Jaguar. *Light examples:* M150, M901, FV438, Striker, and BRDM.[9]

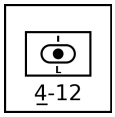


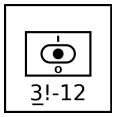


Armored Reconnaissance Platoon: An armored reconnaissance platoon has armored vehicles specialized in scouting and observation and with no offensive weapons more powerful than machine guns. (Vehicles used for reconnaissance that do have significant offensive weapons are typically classified as light tanks or IFVs.) An armored reconnaissance platoon may have light or open armor. *Light examples:* M114, Ferret, and BRDM-2. *Open example:* M20.[10]

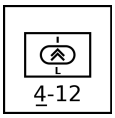


Armored Mortar Platoon: An armored mortar platoon has mortars mounted in armored vehicles (most commonly converted APCs or IFVs). It is typically a battalion-level asset in mechanized-infantry or tank battalions, and its role is indirect fire support. An armored mortar platoon has light armor. *Examples:* M106 and M125.[11]

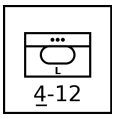




Armored Artillery Battery: An armored artillery battery has guns or howitzers in armored vehicles. It is typically a regiment-, brigade-, or division-level asset, and its role is indirect fire support. An armored artillery battery may have light or open armor. *Light examples:* M109, 2S3, and 2S1.*Open examples:* M7 Priest and AMX-105A.[12]



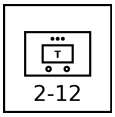
Armored Rocket Artillery Battery: An armored rocket artillery battery is the rocket-armed equivalent of an armored artillery battery. An armored rocket artillery battery has light armor. *Example:* M270.[13]



Armored HQ Platoon: An armored HQ platoon is the HQ unit of a tank, mechanized-infantry, or armored-artillery battalion, regiment, or brigade. It consists of the formation commander, their staff, and their protection detail. It is typically equipped with modified APCs or IFVs. An armored HQ platoon has light armor.

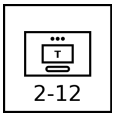
UNARMORED VEHICLE UNITS

Unarmored or soft vehicles provide mobility, but do not provide (or provide only incomplete) armor protection for their crew and weapons. They include armored vehicles whose crew or weapons are exposed. They may be wheeled or tracked.

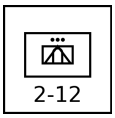


Truck Platoon: A truck platoon has unarmored wheeled vehicles that can transport medium or heavy loads of infantry, towed weapons, and supplies, including fuel and ammunition.

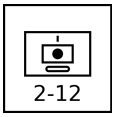
Light Truck Platoon: A light truck platoon has unarmored wheeled vehicles that can transport light loads of infantry, towed weapons, and supplies, including fuel and ammunition. *Examples:* M151 Jeep, HMMWV, Land Rover, Ilitis, and GAZ-69.



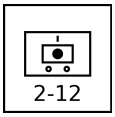
Tracked Carrier Platoon: A tracked carrier platoon is equipped with unarmored tracked vehicles that can transport medium loads of infantry, towed weapons, and supplies, including fuel and ammunition. *Example:* Bv 202.



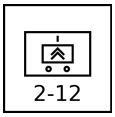
Mobile Anti-Tank Missile Platoon: A mobile anti-tank missile platoon is equipped with anti-tank missiles mounted on unarmored light trucks. *Examples:* M151 TOW and M966 HMMWV TOW.



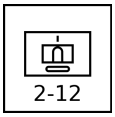
Tracked Artillery Battery: A tracked artillery battery has guns or howitzers mounted on tracked vehicles that provide mobility, but give only limited protection from small arms or artillery shrapnel. It is typically a regiment-, brigade-, or division-level asset, and its role is indirect fire support. *Examples:* M107, M110, 2S4, and 2S7.[14]



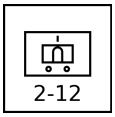
Mobile Artillery Battery: A mobile artillery battery has howitzers mounted on wheeled vehicles that provide mobility, but give only limited protection from small arms or artillery shrapnel. It is typically a regiment-, brigade-, or division-level asset, and its role is indirect fire support. *Examples:* CAESAR and Archer.[15]



Mobile Rocket Artillery Battery: A mobile rocket artillery battery is the rocket-armed equivalent of a mobile artillery battery. *Examples:* M142 HIMARS and BM-21 Grad.[16]



Tracked Missile Artillery Battery: A tracked rocket artillery battery is the rocket-armed equivalent of a mobile tracked battery. *Example:* M752 Lance.[17]



Mobile Missile Artillery Battery: A mobile missile artillery battery is the missile-armed equivalent of a mobile artillery battery. *Examples:* M1003 Pershing II and OTR-21 Tocha (SS-21).[18]

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Mobile HQ Platoon: A mobile HQ platoon is typically the HQ unit of a battalion, regiment, or brigade other than those of infantry, tank, mechanized infantry, or armored artillery. It is equipped with trucks, and its personnel are not specialized in fighting as infantry.

TOWED UNITS

Towed weapons provides neither inherent mobility nor protection. However, they are light and cheap compared to similar vehicle-mounted weapons. Towed weapons is typically used away from direct combat and provided with appropriate ground or helicopter transport.

4-12

Towed Anti-Tank Gun Battery: A towed anti-tank gun battery has anti-tank guns on towed mounts with at most only very limited protection. It is often a regimental-, brigade-, or divisional-level asset. *Example:* 2A18 (T-12) and 2A45 (Sprut).

3-12

Towed Mortar Platoon: A towed mortar platoon has 120 to 160 mm mortars on unprotected, towed mounts. It is often a regimental- or brigade- asset. Smaller infantry mortars are found in infantry mortar platoons. *Example:* M1943.

4-18

Towed Artillery Battery: A towed artillery battery has guns or howitzers on towed mounts with at most only very limited protection. It is often a regimental-, brigade-, or divisional-level asset. *Examples:* M119, M198, Light Gun, and 2A18 (D-30).

Ground-Combat and Transport Unit Table

Type	Size[a]	Defense Strength	Sighting Range	Mobility	VPs 3D/2D/D	AAA Class[b]
Infantry	Platoon	4	6	U	5/3/2	B2
Infantry Weapons	Platoon	3	6	U	6/4/2	B2
Infantry Mortar	Platoon	3	6	U	6/4/2	B2
Infantry FAC	Squad	5	6	U	–/–/8	—
Infantry SAM	Squad	5	6	U		—
Infantry Heavy SAM	Squad	5	6	U		—
Infantry HQ	Platoon	4	6	U	10/7/3	B2
Heavy Tank	Platoon	8	12	U	12/8/4	B3
Medium Tank	Platoon	6	12	U	10/7/3	B3
Light Tank	Platoon	4	12	U	6/4/2	B3
Open Tank	Platoon	3!	12	U	5/3/2	B3
Medium IFV	Platoon	6	12	U	10/7/3	B3
Light IFV	Platoon	4	12	U	6/4/2	B3
Heavy APC	Platoon	8	12	U	10/7/3	B3
Medium APC	Platoon	6	12	U	8/5/3	B3
Light APC	Platoon	4	12	U	6/4/2	B3
Open APC	Platoon	3!	12	U	5/3/2	B3
Light Armored Truck	Platoon	4	12	G/P	6/4/2	B3
Open Armored Truck	Platoon	3!	12	G/P	5/3/2	B3
Medium Armored Anti-Tank Missile	Platoon	6	12	U	12/8/4	B3
Light Armored Anti-Tank Missile	Platoon	4	12	U	10/7/3	B3
Light Armored Reconnaissance	Platoon	4	12	U	6/4/2	B3
Open Armored Reconnaissance	Platoon	3!	12	U	4/3/1	B3
Light Armored Mortar	Platoon	4	12	U	10/7/3	B3
Light Armored Artillery	Battery	4	12	U	12/8/4	B3
Open Armored Artillery	Battery	3!	12	U	12/8/4	B3
Light Armored Rocket Artillery	Battery	4	12	U	12/8/4	B3
Light Armored HQ	Platoon	4	12	U	12/8/4	B3
Truck	Platoon	2	12	G/P	4/3/1	—
Light Truck	Platoon	2	12	G/P	4/3/1	—
Tracked Carrier	Platoon	2	12	U	4/3/1	—
Mobile Anti-Tank Missile	Platoon	2	12	G/P	8/5/3	—
Mobile Artillery	Battery	2	12	G/P	9/6/3	—
Tracked Artillery	Battery	2	12	U	10/7/3	—
Mobile Rocket Artillery	Battery	2	12	G/P	9/6/3	—
Tracked Missile Artillery	Battery	2	12	U	10/7/3	—
Mobile Missile Artillery	Battery	2	12	G/P	9/6/3	—
Mobile HQ	Platoon	2	12	G/P	10/7/3	—
Towed Mortar	Platoon	3	12	T	7/5/2	—
Towed Anti-Tank Gun	Battery	4	12	T	8/5/3	—
Towed Artillery	Battery	4	18	T	8/5/3	—

[a] Platoons and batteries are available as sections and squads with corresponding modifications to their damage resiliences and VPs. [b] Platoons with AAA class of B2 and B3 can employ barrage fire to altitudes of 2 and 3 levels, respectively.

Vehicle Types Table

Heavy Tanks

Al-Khalid/VT-1A	PK/CN	2001
Ariete	IT	1995
Arjun	IN	2004
Challenger 1	GB	1983
Challenger 2	GB	1998
Chieftain	GB	1967
Conqueror	GB	1955
IS-2	SU	1944
IS-3	SU	1945
K1	KR	1987
K2 Black Panther	KR	2014
Leclerc	FR	1992
Leopard 2	DE	1979
M-84	YU	1985
M1 Abrams	US	1980
Merkava 1	IL	1979
Merkava 2	IL	1983
Merkava 3	IL	1989
Merkava 4	IL	2004
PT-91	PL	1995
Stridsvagn 103	SE	1967
T-10	SU	1954
T-14 Armata	RU	2024
T-64	SU	1966
T-72	SU	1973
T-80	SU	1976
T-84	UA	1999
T-90	RU	1992
Type 10	JP	2012
Type 90	JP	1990
Type 99	CN	2001

Medium Tanks

AMX-30	FR	1966
ASU-85	SU	1959
Centurion	GB	1945
Comet	GB	1944
Cromwell	GB	1944
ISU-122	SU	1944
ISU-152	SU	1943
Kanonen JPz	DE	1965
Leopard 1	DE	1965
M-50 Sherman	IL/FR	1956
M-51 Sherman	IL/FR	1961
M103	US	1957
M26 Pershing	US	1944
M4 Sherman	US	1942
M46 Patton	US	1949
M47 Patton	US	1951
M48 Patton	US	1952
M60	US	1959
Panzer 61	CH	1965
Panzer 68	CH	1971
SU-100	SU	1944
T-34	SU	1940
T-54	SU	1955
T-55	SU	1958
T-62	SU	1961
TAM	AR	1983
Type 59	CN	1959
Type 61	JP	1961
Type 69	CN	1972
Type 74	JP	1975
Type 79	CN	1984
Type 85	CN	1988
Type 88	CN	1988
Type 96	CN	1997
Type 15	CN	2018
Vickers MBT	GB	1965
Vijayanta	IN/GB	1965

Light Tanks

AMX-10 RC	FR	1981
AMX-13	FR	1953
Fox	GB	1975
Commando (90 mm)	US	1964
LAV I Cougar	CA	1976
LAV II Coyote	CA	1990
M8/Greyhound	US	1943
M24 Chaffee	US	1944
M41 Walker Bulldog	US	1953
M113 C&V with 25 mm	US	1974
M114A2	US	1969
M551 Sheridan	US	1969
Panhard AML	FR	1961
Panhard EBR	FR	1951
PT-76	SU	1951
Saladin	GB	1958
Scimitar	GB	1974
Scorpion	GB	1973
SpPz Luchs	DE	1975
Stryker MGS	US/CA	2006
Strv 74	SE	1958
Type 62	CN	1961
Type 63	CN	1963
Type 05 AAV	CN	2005
Type 08 ARV	CN	2008

Open Tanks

ASU-57	SU	1951
SU-76	SU	1943

Medium IFVs

BMP-3	SU	1987
Boxer IFV	NL/DE	2011
Strf 9040/CV90	SE	1994
LAV III (add-on-armor)	CA	2009
M2A2/3/4 Bradley	US	1988
M3A2/3/4 Bradley	US	1988
SPz Marder	DE	1971
SPz Puma	DE	2015

Light IFVs

AIFV	US	1979
AMX-10P	FR	1973
BMP-1	SU	1966
BMP-2	SU	1980
Commando (20 mm)	US	1963
FV432 RARDEN	GB	1975
LAV III	CA	1999
LAV-25	CA	1983
M2 and M2A1 Bradley	US	1981
M3 and M3A1 Bradley	US	1981
M113A1 FSC	AU	1972
Piranha IFV	CH	1974
Pbv 301	SE	1961
Pbv 302	SE	1966
SPz Kurz	FR	1959
SPz Lang HS.30	CH	1960
Stryker ICVD	US	2018
Type 86	CN	1992
Type 92	CN	1992
Type 05 IFV	CN	2005
Type 08 IFV	CN	2008
VCBI	FR	2008
Warrior	GB	1987
YPR-765	NL	1977

Heavy APCs

Achzarit	IL	1988
Nakpadon	IL	1998
Namer	IL	2008
Puma	IL	1993

Medium APCs

Boxer APC	NL/DE	2011
Nagmachon	IL	2000
Nagmashot	IL	1990

Light APCs

Bronco/Warthog	SG	2001
BvS10/Viking	SE	2005
BTR-50PK	SU	1958
BTR-60PA/PAI/PB	SU	1965
BTR-70	SU	1972
BTR-80	SU	1986
Commando	US	1963
FV432	GB	1963
LAV I Grizzly	CA	1976
LAV II Bison	CA	1990
LVTP-5	US	1956
LVTP-7 (AAV)	US	1972
M59	US	1954
M75	US	1952
M113	US	1960
OT-64 SKOT	CS/PL	1963
OT-810 half-track	CS	1958
Piranha APC	CH	1974
Saracen	GB	1952
Spartan	GB	1978
Stryker ICV	US/CA	2002
TPz Fuchs	DE	1979
Type 63	CN	1978
Type 77	CN	1978
Type 89	CN	1978
Type 08 APC	CN	2008
VAB	FR	1979
YP-408	NL	1964

Open APCs

AT-P	SU	1954
BTR-50P	SU	1952
BTR-60P	SU	1959
M3 half-track	US	1941

Light Armored Trucks

BTR-152K	SU	1959
Saxon	GB	1983

Open Armored Trucks

BTR-40	SU	1950
BTR-152A	SU	1950
SKP/VKP	SE	1944
SKPF/VKPF	SE	1956

Medium Armored Anti-Tank

Raketen JPz 1	DE	1961
Raketen JPz 2	DE	1967
Jaguar 1	DE	1978
Jaguar 2	DE	1983

Light Armored Anti-Tank

BRDM-2 ATGM	SU	1973
BRDM-3 ATGM	SU	1973
Fennek MRAT	DE/NL	2003
FV438 Swingfire	GB	1971
M150	US	1973
M901 ITV	US	1979
Striker	GB	1976
Stryker ATGM	US/CA	2003
VAB HOT	FR	1978
YP-408 PRAT	NL	1976
YPR-765 PRAT	NL	1980

Light Armored Reconnaissance

BRDM-2	SU	1973
Fennek	DE/NL	2003
Ferret	GB	1952
M113 C&V/Lynx	US	1966
M114	US	1962

Open Armored Reconnaissance

M20	US	1943

Light Armored Mortar

FV432 with 81 mm	GB	1963
M106	US	1964
M125	US	1964
M1064	US	1990
Stryker MCV	US/CA	2005

Light Armored Artillery

2S1 Gvozdika	SU	1972
2S3 Akatsiya	SU	1971
2S19 Msta-S	SU	1989
AMX-30 AuF1	FR	1977
AS-90	GB	1992
Bkan 1	SE	1967
FV433 Abbot	GB	1965
L-33 Ro'Em	IL	1973
M52	US	1955
M108	US	1962
M109	US	1963
Panzerhaubitze 2000	DE	1998

Open Armored Artillery

AMX-105A	FR	1958
M-50	IL	1962
M7 Priest	US	1942
M40	US	1945
M44	US	1954

Light Armored Rocket Artillery

M270 MLRS	US	1983
BM-1	RU	1988

Light Truck

GAZ-69	SU	1953
HMMWV	US	1985
Ilitis	GE	1978
Landrover	UK	1949
M38 1/4-ton truck	US	1949
M151 1/4-ton truck	US	1960
UAZ-469	SU	1971
Willys MB 1/4-ton truck	US	1941

Tracked Carrier

AT-T/S/L	SU	1950
Bv 202	SE	1964
Bv 206	SE	1980
BvS10/Beowulf	SE	2024
M548	US	1960
Sisu Nasu	FI	1986

Mobile Anti-Tank Missile

M151 TOW	US	1970
M966 HMMWV TOW	US	1985

Mobile Artillery

Archer	SE	2016
CAESAR	FR	2004

Tracked Artillery

2S4 Tyulpan	SU	1975
2S7 Pion	SU	1976
M41	US	1942
M107	US	1962
M110	US	1963
AMX-13 F3	FR	1962

Mobile Rocket Artillery

BM-8	SU	1941
BM-13	SU	1941
BM-31	SU	1944
BM-24	SU	1947
BM-14	SU	1952
BM-21 Grad	SU	1963
BM-27 Uragan	SU	1975
BM-30 Smerch	SU	1989
M142 HIMARS	US	2005

Tracked Missile Artillery

M752 Lance	US	1972
AMX-30 Pluton	FR	1974

Mobile Missile Artillery

9K52 Luna-M (Frog-7)	SU	1962
Corporal	US	1954
Sergeant	US	1962
M34/M50 Honest John	US	1953
M74 Redstone	US	1958
M790 Pershing 1a	US	1962
M1003 Pershing II	US	1983
OTR-21 Tochka (SS-21)	SU	1975
RSD-10 Pioneer (SS-20)	SU	1976

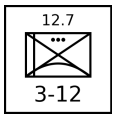
AAA UNITS

AAA units use guns to destroy attacking aircraft, but also have a secondary ground-combat capability. Some also have SAMs, giving a layered capacity. AAA units are presented below with their counters, ordered first by caliber and then by year of introduction.

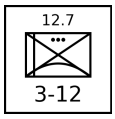
The AAA Unit Table summarizes the properties of AAA units. As well as the basic properties common to all ground units, the table gives: the AAA class (light, medium, or heavy); the maximum altitude; the limits of short, medium, and long range in hexes; the hit rolls at short, medium, and long range; the damage rating; whether the unit has an all-weather or ranging-only fire-control radar and its frequency band; whether it has night infrared sights; and whether it is also equipped with a SAM system. If the unit has a SAM, this will be described in detail in the section on SAM launcher units. [19]

All AAA platoons and batteries are also available as sections (but not squads), with damage resiliences of 2, hit rolls reduced by 1, and VPs corresponding to a platoon or battery that has taken 1D damage.

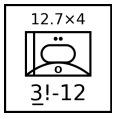
LIGHT AAA UNITS



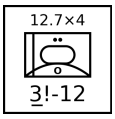
M2 Platoon: The Browning M2 .50 cal (12.7 mm) heavy machine gun is normally found on a low tripod for ground combat or mounted on armored fighting vehicles, but here is used on a tall dual-purpose mount suitable for air defense. A platoon has six guns. The M2 gun was introduced in 1933 and has been used by United States forces and their allies in numerous conflicts since then.[20]



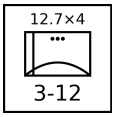
DShK-38 Platoon: The DShK-38 12.7 mm heavy machine gun is most commonly found on a low, wheeled mount for ground combat or mounted on armored fighting vehicles, but here is used on a tall tripod suitable for air defense. It is nicknamed the Dushka (darling) or Dochka (daughter). A platoon has six guns. The DShK-38 entered service with the Soviet Army in 1938 and saw action in World War II and numerous conflicts since then, including the Korean and Vietnam Wars, in which the Soviet Union participated or provided arms. In particular, it was the main air-defense gun for Chinese and North Korean field units in the early part of the Korean War.[21]



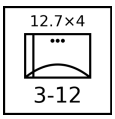
M16 Section: The M16 Multiple Gun Motor Carriage has an M45 turret with four M2 .50 cal (12.7 mm) heavy machine guns mounted on a modified M3 half-track. A section has two vehicles. The M16 is a development on the M13 and M14 Multiple Gun Motor Carriages, which have two M2 heavy machine guns in a similar turret. As a single M2 has an insufficient rate of fire to be effective even against WW2 aircraft, first two and then four were used on a single mount. The M16 was used by United States forces during World War II. It was also used by US, Commonwealth, and South Korean forces in the Korean War, primarily in an anti-infantry role. It was largely withdrawn form US services in late 1951 and officially retired in 1958. Israeli forces employed it during the 1956 Arab-Israeli War. Other post-WW2 users include Belgium, France, West Germany, Japan, Netherlands, Philippines, Portugal, and Thailand[22]



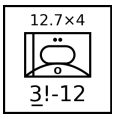
M17 Section: The M17 Multiple Gun Motor Carriage is the equivalent of the M16 Multiple Gun Motor Carriage, but based on the slightly different M5 half-track. A section has two vehicles. The M17 was supplied to the Soviet Union and entered service in 1944. It was presumably replaced by the BTR-40A.[23]



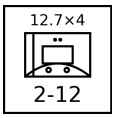
M55 Platoon: The M55 has an M45 turret with four M2 .50 cal (12.7 mm) heavy machine guns (the same turret as used in the M16) mounted on a towed carriage. A platoon has four gun mounts. The M55 was used by United States forces during the Korean War and by Pakistani forces during the 1965 India-Pakistan War.[24]



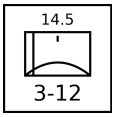
Vz.53 Platoon: The Czechoslovak Vz.53 has four Vz.38/46 12.7 mm machine guns (copies of the Soviet DShK) on a light, wheeled mount. It is sometimes referred to as the M53. A platoon has six guns. It entered service with the Czechoslovak Army in 1953, and was exported to Cuba, Egypt (from 1955), Iraq, Syria, and Vietnam. Cuba subsequently exported some to Grenada. It saw action with the Egyptian Army in the 1956 War, with Cuba in the Bay of Pigs Invasion in 1961, in the Vietnam War, in the Nicaraguan Revolution, with the Grenadian Army during the invasion by US force in 1983[25]

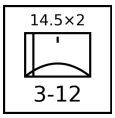


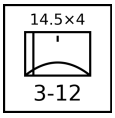
BTR-152A Vz.53 Section: The BTR-152A Vz.53 has a Czechoslovak Vz.53 gun mount with four 12.7 mm Vz.38/46 machine guns in the open, six-wheeled Soviet BTR-152 APC. The gun mount is also referred to as the M53. A section has two vehicles. It served only in the Egyptian Army, and saw action in the 1956 War.[26]



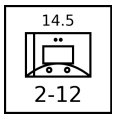
Mobile M45 Section: The mounted M45 has the M45 turret with four M2 .50 cal (12.7 mm) heavy machine guns (the same turret as used in the M16) removed from an M55 trailer and mounted on a truck. During the Vietnam War, in 1967, US forces improvised these *gun trucks* for convoy security[27]

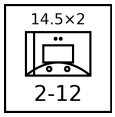


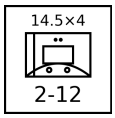




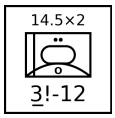
ZPU-1/2/4 Battery: The ZPU-1/2/4 have one, two, or four 14.5 mm heavy machine guns in open mounts on towed carriages. A battery has six gun mounts. The ZPU-1/2/4 was introduced in 1949 and saw extensive service with Soviet or Soviet-supported forces in many subsequent conflicts, including with communist forces in the Korean War and Vietnam War and with Arab forces in the various Arab-Israeli Wars from 1956 onwards.[28]



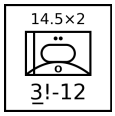


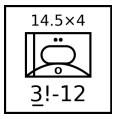


Mobile ZPU-1/2/4 Section: The mobile ZPU-1/2/4 is a light truck with a ZPU-1/2/4 gun mount. A section has two vehicles. Many users of the ZPU-1/2/4 adapt it in this way for increased mobility, and like the towed versions, these mounted versions have seen combat in many conflicts. Formal conversions tend to be used for air-defense, but informal conversions or *technicals* are mainly use for ground combat, serving as light cavalry or infantry support.[29]

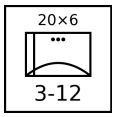


BTR-40A Section: The BTR-40A has a twin 14.5 mm ZPU-2 turret in the open, four-wheeled BTR-40 APC. A section has two vehicles. The BTR-40A entered service with the Soviet Army in 1950, presumably replacing the M17. It was also used by East Germany and Cuba.[30]

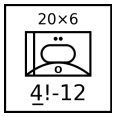




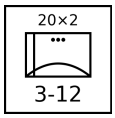
BTR-152A ZPU-2/4 Section: The BTR-152A has a twin 14.5 mm ZPU-2 or quad 14.5 mm ZPU-4 turret in the open, six-wheeled BTR-152 APC. A section has two vehicles. The BTR-152A entered service with the Soviet Army in 1951, presumably replacing the BTR-40A. The BTR-152D/E are similar, but based on the improved BTR-152V/V1 variants. Other users include Ethiopia in the Ogaden War and Egypt.[31]



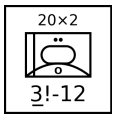
M167 Platoon: The M167 Vulcan Air Defense System (VADS) has a 20 mm six-barreled Vulcan gun in an open mount on a towed carriage. A platoon has four gun mounts. The fire-control system has optical sights and a range-only radar. The M167 was introduced into service with the US Army in 1967, principally for airfield defense, and subsequently served with other US allies.[32]



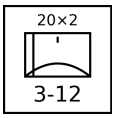
M163 Section: The M163 Vulcan Air Defense System (VADS) has a 20 mm six-barreled Vulcan gun in a lightly armored but open-topped turret on an adapted M113 APC. A section has two vehicles. The fire-control system has optical sights and a range-only radar. The M163 VADS was introduced into service with the US Army in 1968 and has since been adopted by other US allies. The M163 was used by US forces in the Vietnam War, the 1989 Invasion of Panama, and the 1991 Gulf War. It was also used by the Israel Army in the 1982 Lebanon War and in 2002 during the Second Intifada. From 1988 in the US Army, each M163 was typically issued with a FIM-92 Stinger launcher with two rounds, so each M163 section effectively carried a pair of FIM-92 squads.[33]



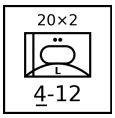
TCM-20 Platoon: The TCM-20 is an Israeli adaptation of the M55, with two Hispano-Suiza 20mm guns from obsolete aircraft replacing the four .50 cal machine guns. As with the M55, the fire-control system is entirely optical. A platoon has four gun mounts. The TCM-20 was employed by Israeli forces from 1970 and subsequently saw combat in the 1973 Arab-Israeli War and the 1982 Lebanon War.[34]



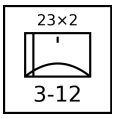
M3 TCM-20 Section: The M3 TCM-20 has a TCM-20 turret Hispano-Suiza 20mm guns mounted in an adapted M3 half-track, like the earlier M16. A section has two vehicles. The M3 TCM-20 was used by Israeli forces in the 1973 Arab-Israeli War and the 1982 Lebanon War.[35]



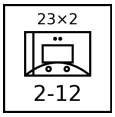
Twin Rh-202 Battery: The Rheinmetall Rh-202 has a single or twin 20mm gun in an open mount on a towed carriage. The fire-control system is entirely optical. The most common version is the twin mount. A battery has six gun mounts. The twin version entered service with West German forces in 1972 and was used for airfield defense. Subsequently, it was exported and notably used by the Argentine Air Force in the 1982 South Atlantic War in the defense of BAM Malvinas (Port Stanley Airport) and BAM Condor (Goose Green Airfield).[36]



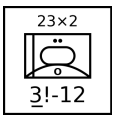
Panhard M3 VDA Section: The Panhard M3 VDA vehicles have an armored turret for twin HS.820 20 mm guns mounted on the Panhard M3 APC. A section has two vehicles. The guns have optical sights, but typically one vehicle in each section or platoon is equipped with a search-and-ranging radar and can share this information automatically with the others. The Panhard M3 VDA served with the armed forces of Côte d'Ivoire, Niger, and the UAE.[37]



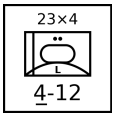
ZU-23 Battery: The ZU-23 has twin 23 mm cannons in an open mount on a towed carriage. It was designed as a more powerful replacement for the ZPU-1/2/4. A battery has six gun mounts. The fire-control system is entirely optical. The ZU-23 was introduced into service by the Soviet Army in 1960. It later served with the armed forces of many Soviet allies, including with North Vietnam in the Vietnam War, with Egypt and Syria in the 1967 and 1973 Wars, with Soviet forces in the Soviet-Afghan War, with Iraq in the Iraq-Iran War, with Syria in the 1982 Lebanon War, and with both sides in the Afghan Civil War.[38]



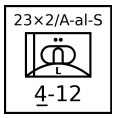
Mobile ZU-23 Section: This is a section of two light trucks mounting ZU-23 guns. As with the earlier ZPU-1/2/4, many users of the ZU-23 mount it on vehicles, including trucks, tracked carriers, and APCs, for better mobility. Informal conversions are typical used as *technicals* in ground combat.[39]

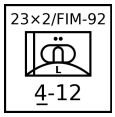


BTR-152A ZU-23 Section: The BTR-152A ZU-23 has a twin 23 mm ZU-23 turret in the open, six-wheeled BTR-152 APC. A section has two vehicles. It entered service with the PLO prior to the 1982 War in Lebanon; the in-service data is a guess.[40]

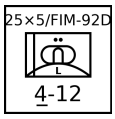


ZSU-23-4 Section: The ZSU-23-4 has four radar-controlled 23 mm guns, similar to those in the ZU-23, mounted in an armored turret on a hull adapted from the PT-76 light tank. A section has two vehicles. The ZSU-23-4 entered service in the Soviet Army in 1965, replacing the ZSU-57-2 and outclassing all NATO anti-aircraft guns at that time. A platoon of four was assigned to the air-defense battery of tank and motor-rifle regiments and later complemented by a platoon of four SA-9 vehicles. It was exported to Soviet allies and saw extensive combat with Arab forces in the 1973 Arab-Israeli War and with Iraqi forces in the Iran-Iraq War and the 1991 Gulf War.[41]

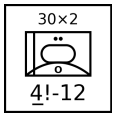




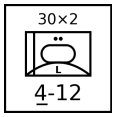
Sinai 23 Section: The Sinai 23 has a Dassault TA20 turret with two 23 mm guns and six Ayn-al-Saqr (similar to the Strela-2M) or Stinger missiles on an M113 chassis. A section has two vehicles. Each battery of two sections operates with one vehicle equipped with an RA 20S search and ranging radar. The Sinai 23 entered service with the Egyptian Army in 1991.[42]



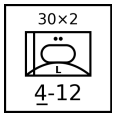
LAV-AD Section: The LAV-AD has a 25 mm GAU-12 rotary cannon and eight FIM-92D Stinger missiles mounted in a turret on the LAV-25 vehicle. A section has two vehicles. The LAV-AD does not have radar, but is equipped with a passive night IR sight. It entered service with the USMC in 1997, but was retired after only a few years.[43]



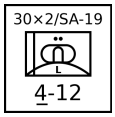
M53/59 Section: The M53/59 has a twin M53 30 mm gun mount on a partially armored truck. The fire-control system is entirely optical. A section has two vehicles. It entered service in the Czech Army in 1959 and was used instead of the ZSU-57-2. It was also exported to Iraq and saw combat in the Iran-Iraq War and the 1991 Gulf War.[44]



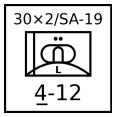
AMX-13DCA Section: The AMX-13DCA is equipped with a pair of radar-controlled 30 mm Hispano-Suiza guns mounted in an armored turret on the hull of an AMX-12 tank. A section has two vehicles. The AMX-13DCA is equipped with the Thomson-CSF Oeil Noir radar. It served only in the French Army, entering service in 1969 and being retired in the 1990s.[45]



AMX-30SA Section: The AMX-30SA is equipped with a pair of radar-controlled 30 mm Hispano-Suiza guns mounted in an armored turret on the hull of an AMX-30 tank. A section has two vehicles. AMX-30SA is an improved version of the AMX-30 DCA and served only in the Saudi Army from 1979.[46]

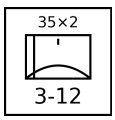


Tunguska Section: The Tunguska has twin radar-controlled 30mm guns and four SA-19 missiles in an armored turret on an armored and tracked hull. A section has two vehicles. The Tunguska was designed to counter the A-10 and AH-64, against which the earlier ZSU-23-4 had weaknesses. It entered service in the Soviet Army in 1984. It has been used by Russia, Ukraine, and Belarus after the break-up of the Soviet Union and has been exported. The Tunguska is referred to as the "ZSU-30-2" in *Air Strike*.[47]

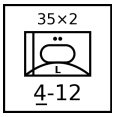


Tunguska-M Section: The Tunguska-M is an improved version of the Tunguska with eight missiles instead of four. A section has two vehicles. It entered service in the Soviet Army in 1990. It has been used by Russia, Ukraine, and Belarus after the break-up of the Soviet Union and has been exported.[48]

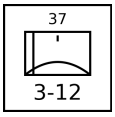
MEDIUM AAA UNITS



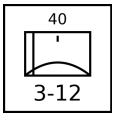
Oerlikon GDF Battery: The Oerlikon GDF has two 35 mm guns in an open mount on a towed carriage. A battery has six gun mounts. The basic fire-control system is optical, but a section is often coupled with the Super Fledermaus or Skyguard fire-control radars. (These radars cannot control a whole battery, only a section.) The Oerlikon GDF entered service in 1963 and has served widely. Notably, it saw combat in the South Atlantic War with the Argentine forces in the defense of BAM Malvinas (Port Stanley Airport) and BAM Condor (Goose Green Airfield).[49]



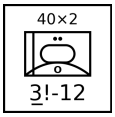
Gepard Section: The Gepard has two radar-controlled 35 mm guns in an armored turret on an adapted Leopard 1 tank hull. A section has two vehicles. The Gepard entered service in 1976 with the West German Army and also served with the Belgian and Dutch armies. After the end of the Cold War, they were retired by these armies, and many were sold on to other countries.[50]



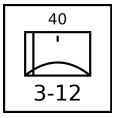
61-K Battery: The 61-K (M1939) has a single 37 mm gun in an open mount on a towed carriage. A battery has six gun mounts. The fire-control system is entirely optical, and (as an exception to the normal rule allowing medium AAA units to employ external FCRs) the 61-K cannot be coupled to an external fire-control radar. After entering service in 1939 with the Soviet Army, it served in WW2 and after until replaced by the S-60 57 mm gun. It also served with many Soviet allies, and in particular with communist forces in the Korean and Vietnam Wars. The 61-K is referred to as the "M-38" in *Air Strike*, *Eagles of the Gulf*, and *The Speed of Heat*.[51]



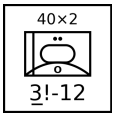
Bofors L/60 Battery: The Bofors L/60 has a single 40 mm gun in an open mount on a towed carriage. A battery has six gun mounts. The fire-control system is entirely optical. The Bofors L/60 entered service in the 1940s and was widely used by Allied forces during WW2, both on land and sea. After the war, many continued to see service, although it was increasingly inadequate against faster jet aircraft and in many cases was replaced by the new L/70 model.[52]



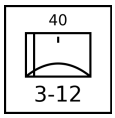
M19 Section: The M19 Multiple Gun Motor Carriage has two Bofors 40 mm L/60 guns in a partially armored mount on a tracked chassis based on that of the M24 light tank. A section has two vehicles. The fire-control system is entirely optical. The M19 entered service in 1944 with the US Army, but did not see combat in WW2. It was used in the Korean War, mostly in against ground targets, but was retired in 1953 and replaced by the M42. It served in the Netherlands Army from 1951 to 1978.[53]



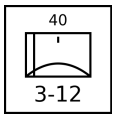
Bofors L/70 Battery: The Bofors L/70 has a single 40 mm gun in an open mount on a towed carriage. A battery has six gun mounts. Between the 1930s and the end of WW2, aircraft speed increased substantially, and so the engagement time of the Bofors L/60 became inadequate. The new L/70 fired a lighter shell at a higher velocity and achieved a significant improvement in range and engagement time. The basic fire-control system is optical, but various external fire-control radars can be used with it, including the Super Fledermaus and Flycatcher systems. NATO forces and others adopted it widely starting in 1952. Israel used batteries with the Super Fledermaus FCR for point-defense in the 1967 war.[54]



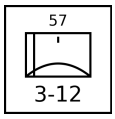
M42 Section: The M42 Duster has two Bofors 40 mm L/60 guns in a partially armored mount on a tracked chassis based on that of the M41 light tank. A section has two vehicles. The fire-control system is entirely optical. The M42 entered service in 1953 with the US Army and US Marine Corps, replacing the M19, saw combat in the Vietnam War against ground targets, and served at least with reserve units until 1988. It was also used by Austria, Greece, Japan, Jordan, Lebanon, Pakistan, Taiwan, Thailand, Tunisia, Turkey, Venezuela, Vietnam (South and North), and West Germany.[55]



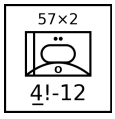
Bofors L/70 BOFI: The BOFI version of the Bofors L/70 adds modern sights, a laser rangefinder, and proximity-fuzed shells. A battery has six gun mounts. The BOFI is used by the armed forces of Brazil.[56]



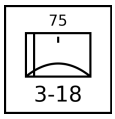
Bofors L/70 BOFI-R Battery: The BOFI-R version of the Bofors L/70 also adds an integrated fire-control radar to the BOFI version. A battery has six gun mounts.[57]



S-60 Battery: The S-60 is a Soviet single 57 mm gun on an open mount on a towed carriage. A battery has six gun mounts. The basic fire-control system is optical, but the gun is often used with the SON-9 (Fire Can) radar. The S-60 entered service in 1950 with Soviet forces, replacing the 61-K 73 mm gun. It was exported to many Soviet allies and saw combat with Arab forces in the 1967 and 1973 Arab-Israeli War, with communist forces in the Vietnam War, and with Iraqi forces in the Iran-Iraq War and the Gulf War. However, it was not used by communist forces in the Korean War. When paired with the SON-9 (Fire Can) radar, it was considered one of the more dangerous AAA systems faced by US aircraft in Vietnam.[58]

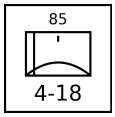


ZSU-57-2 Section: The ZSU-57-2 has two 57 mm cannons in a lightly armored but open-topped turret on an armored and tracked hull. A section has two vehicles. The cannons are similar to the S-60 cannons and the fire-control system is optical. The ZSU-57-2 entered service in the Soviet Army in 1955 and replaced the BTR-40A and BTR-152A self-propelled 14.5 mm AA guns in the AA batteries of tank regiments. It was in turn replaced by the ZSU-23-4 from 1965. The ZSU-57-2 also served widely with Soviet allies and saw combat with Egypt and Syria in the 1967 and 1973 Wars, with Syria in the 1982 Lebanon War, with North Vietnam in the 1972 Easter Offensive and the 1975 Spring Offensive, and with various factions in the Yugoslav Wars in the 1990s.[59]

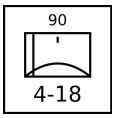


M51 Battery: The M51 Skysweeper has a single 75 mm gun in an open mount in a towed carriage. A battery has four mounts. The mount incorporates the T38 radar fire-control system and the gun has a relatively high rate of fire of 45 rounds per minute. Proximity-fuzed shells are available. The M51 entered service with the US Army in 1951 and was used for defense of strategic locations in the USA and in Europe, but was retired in 1959. It also served with Greece, Japan, and Turkey, with the last examples being retired until the mid-1970s. [60]

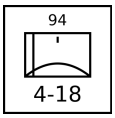
HEAVY AAA UNITS



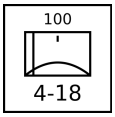
KS-12 Battery: The KS-12 (M1939 52-K) and the very similar KS-12A (M1944 KS-1) have a single 85 mm gun in an open mount on a towed carriage. A battery has six gun mounts. The rate of fire of the KS-12 is 10 rounds per minute. Its fire-control system is optical, but the guns are often paired with the SON-9 (Fire Can) radar. The KS-12 entered service with Soviet forces in 1939. After WW2, it began to be replaced in Soviet service by the KS-19 100 mm and KS-30 130 mm guns, but was widely used by Soviet allies and saw combat with communist forces in the Korean War and the Vietnam War.[61]



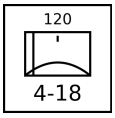
90 mm M2 Battery: The 90 mm M2 has a single 90 mm gun on a towed carriage. A battery has four guns. The fire control system of the M2 is optical, but an SCR-584 FCR-A normally controls each battery. The mechanical loading system permits a rate of fire of 24 rounds per minute. Proximity-fuzed shells are available. The M2 entered service with the US Army in 1943, following on from the similar M1 in 1940, and served as the main US heavy AA gun in WW2 and the Korean War. It was retired in the mid 1950s. It also served with Canadian (until 1960), French, and Turkish forces.[62]



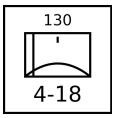
QF 3.7 inch Battery: The QF 3.7 inch has a single 94 mm gun on a towed carriage, but typically is used from a prepared firing site. The fire control system is optical, but an SCR-584 FCR-A normally controls each battery. In the final Mk VI version, the mechanical loading system permits a rate of fire of 20 rounds per minute. Proximity-fuzed shells are available. The QF 3.7 inch entered service with the British Army in 1937, served as its principal heavy AA gun during WW2 and beyond, and was withdrawn in 1957. It also served with Australia, Belgium, Canada, Sri Lanka, Cyprus, India, Ireland, Israel, Malta, Nepal, New Zealand, Pakistan, South Africa, Yugoslavia, and Portugal.[63]



KS-19 Battery: The KS-19 has a single 100 mm gun in an open mount on a towed carriage. A battery has six gun mounts. The fire-control system of the KS-19 is optical, but like other contemporary Soviet guns, it was often paired with the SON-9 (Fire Can) radar. The KS-19 entered service in 1948 with Soviet forces as a replacement for the KS-12 85 mm gun. It also served with many Soviet allies and saw combat with communist forces in the Korean War and Vietnam War and Iraqi forces in the Iran-Iraq and Gulf Wars.[64]



120 mm M1 Battery: The M1 has a single 120 mm gun in an open mount on a towed carriage, but typically is used from a prepared firing site. A battery has four guns. The fire control system of the M1 is optical, but an SCR-584 FCR-A normally controls each battery. The rate of fire is 12 rounds per minute. Proximity-fuzed shells are available. The M1 entered service with the US Army in 1944, and though deployed to the Pacific Theater did not engage on enemy aircraft. After the war, it was widely deployed to defend of strategic targets against Soviet Tu-4 bombers, but was withdrawn by 1960. It also served with Canadian forces.[65]



KS-30 Battery: The KS-30 has a single 120 mm gun in an open mount on a towed carriage. A battery has six gun mounts. Development of the KS-30 started after WW2, and it has features adopted from captured German guns as well as British QF 3.7 inch guns. The fire-control system is optical, but like other contemporary Soviet guns, it was often paired with the SON-9 (Fire Can) radar. The rate of fire is 12 rounds per minute. The KS-30 entered service in 1955 with Soviet forces, was used to defend strategic sites, and served until 1962. It was not widely exported, but also served with the North Vietnamese forces in the Vietnam War and Iraqi forces in the Iran-Iraq and Gulf Wars.[66]

AAA Units Table

	Type	Country	Year	Gun Size[a]	Defense Strength	Sighting Range	Mobility	VPs 3D/2D/D	AAA Class	Altitude	Range S	Range M	Range L	Hit S	Hit M	Hit L	Damage Rating	FCR	Night IR Sights	SAM				
M2	12.7 mm	US	1933	Platoon	3	12	U	3/2/1	L		5	2	3		4		3	2		1	1	—	—	—
DShK-38	12.7 mm	SU	1938	Platoon	3	12	U	3/2/1	L		5	2	3		4		3	2		1	1	—	—	—
M16	12.7 mm × 4	US	1944	Section	3!	12	U	6/3	L		5	2	3		4		4	3		2	2	—	—	—
M17	12.7 mm × 4	US	1944	Section	3!	12	U	6/3	L		5	2	3		4		4	3		2	2	—	—	—
M55	12.7 mm × 4	US	1945	Platoon	3	12	T	4/3/1	L		5	2	3		4		5	4		3	2	—	—	—
Vz.53	12.7 mm × 4	CS	1953	Platoon	3	12	T	4/3/1	L		5	2	3		4		5	4		3	2	—	—	—
BTR-152A Vz.53	12.7 mm × 4	CS	1955	Section	3!	12	G/P	6/3	L		5	2	3		4		4	3		2	2	—	—	—
Mobile M45	12.7 mm × 4	US	1967	Section	2	12	G/P	4/2	L		5	2	3		4		4	3		2	2	—	—	—
ZPU-1	14.5 mm	SU	1949	Battery	3	12	T	3/2/1	L		6	2	3		5		3	2		1	1	—	—	—
ZPU-2	14.5 mm × 2	SU	1949	Battery	3	12	T	4/3/1	L		6	2	3		5		4	3		2	1	—	—	—
ZPU-4	14.5 mm × 4	SU	1949	Battery	3	12	T	5/3/2	L		6	2	3		5		5	4		3	2	—	—	—
Mobile ZPU-1	14.5 mm	SU	1949	Section	2	12	G/P	2/1	L		6	2	3		5		2	1		0	1	—	—	—
Mobile ZPU-2	14.5 mm × 2	SU	1949	Section	2	12	G/P	3/2	L		6	2	3		5		3	2		1	1	—	—	—
Mobile ZPU-4	14.5 mm × 4	SU	1949	Section	2	12	G/P	4/2	L		6	2	3		5		4	3		2	2	—	—	—
BTR-40A	14.5 mm × 2	SU	1950	Section	3!	12	G/P	4/2	L		6	2	3		5		3	2		1	1	—	—	—
BTR-152A ZPU-2	14.5 mm × 2	SU	1951	Section	3!	12	G/P	4/2	L		6	2	3		5		3	2		1	1	—	—	—
BTR-152A ZPU-4	14.5 mm × 4	SU	1951	Section	3!	12	G/P	5/3	L		6	2	3		5		4	3		2	2	—	—	—
M167	20 mm × 6	US	1967	Platoon	3	12	T	7/5/2	L		8	2	4		6		6	5		4	3	R/HF	—	—
M163	20 mm × 6	US	1968	Section	4!	12	U	8/4	L		8	2	4		6		5	4		3	3	R/HF	—	—
TCM-20	20 mm × 2	IL	1970	Platoon	3	12	T	5/3/2	L		8	2	4		6		4	4		3	2	—	—	—
M3 TCM-20	20 mm × 2	IL	1970	Section	3!	12	U	6/3	L		8	2	4		6		3	3		2	2	—	—	—
Single Rh-202	20 mm	DE	1972	Battery	3	12	T	4/3/1	L		8	2	4		6		4	3		2	2	—	—	—
Twin Rh-202	20 mm × 2	DE	1972	Battery	3	12	T	5/3/2	L		8	2	4		6		5	4		3	3	—	—	—
Panhard M3 VDA	20 mm × 2	FR	1975	Section	4	12	U	7/4	L		8	2	4		6		4	4		3	2	R/UF	—	—
ZU-23	23 mm × 2	SU	1960	Battery	3	12	T	6/4/2	L		9	2	5		8		5	4		3	3	—	—	—
Mobile ZU-23	23 mm × 2	SU	1960	Section	2	12	G/P	5/3	L		9	2	5		8		4	3		2	3	—	—	—
ZSU-23-4	23 mm × 4	SU	1965	Section	4	12	U	9/5	L		9	2	5		8		5	4		3	3	W/VF	—	—
BTR-152A ZU-23	23 mm × 2	PS	1975	Section	3!	12	G/P	4/2	L		9	2	5		8		4	3		2	3	—	—	—
Sinai 23 Ayn-al-Saqr	23 mm × 2	EG	1992	Section	4	12	U	10/5	L		9	2	5		8		5	4		3	3	R/UF	—	Ayn-al-Saqr
Sinai 23 Stinger	23 mm × 2	EG	1992	Section	4	12	U	12/6	L		9	2	5		8		5	4		3	3	R/UF	—	FIM-92
LAV-AD	25 mm × 5	US	1997	Section	4	12	U	14/7	L		10	3	6		9		5	4		3	4	—	Y	FIM-92D
M53/59	30 mm × 2	CS	1959	Section	4!	12	G/P	7/4	L		11	4	7		9		3	2		1	4	—	—	—
AMX-13DCA	30 mm × 2	FR	1969	Section	4	12	U	10/5	L		11	4	7		9		5	4		3	3	W/VF	—	—
AMX-30SA	30 mm × 2	FR	1979	Section	4	12	U	10/5	L		11	4	7		9		5	4		3	3	W/VF	—	—
Tunguska	30 mm × 2	SU	1984	Section	4	12	U	20/10	L		11	4	7		9		5	4		3	4	W/VF	—	SA-19
Tunguska-M	30 mm × 2	SU	1990	Section	4	12	U	22/11	L		11	4	7		9		5	4		3	4	W/VF	—	SA-19

AAA Units Table

	Country	Year	Type	Gun Size[α]	Defense Strength	Sighting Range	Mobility	VPs 3D/2D/D	AAA Class	Altitude	Range S	Range M	Range L	Hit S	Hit M	Hit L	Damage Rating	FCR	Night IR Sights	SAM				
Oerlikon GDF	CH	1963	35 mm × 2	Battery	3	12	T	7/5/2	M	12	4	8	10		4	3		2	4	—	—	—
Oerlikon GDF	CH	1963	35 mm × 2	Section	3	12	T	5/2	M	12	4	8	10		3	2		1	4	—	—	—
Gepard	DE	1976	35 mm × 2	Section	4	12	U	11/6	M	12	4	8	10		5	4		3	4	W/VF	—	—
61-K	SU	1939	37 mm	Battery	3	12	T	5/3/2	M	12	4	8	11		3	2		1	4	—	—	—
Bofors L/60	SE	1935	40 mm	Battery	3	12	T	5/3/2	M	15	4	8	12		2	2		1	4	—	—	—
M19	US	1944	40 mm × 2	Section	3!	12	U	6/3	M	15	4	8	12		2	2		1	4	—	—	—
Bofors L/70	SE	1952	40 mm	Battery	3	12	T	6/4/2	M	15	4	8	12		3	3		2	4	—	—	—
M42	US	1953	40 mm × 2	Section	3!	12	U	6/3	M	15	4	8	12		2	2		1	4	—	—	—
Bofors L/70 BOFI	SE		40 mm	Battery	3	12	T	8/5/3	M	15	4	8	12		5	5		4	4	—	—	—
Bofors L/70 BOFI-R	SE		40 mm	Battery	3	12	T	9/6/3	M	15	4	8	12		6	6		5	4	W/VF	—	—
S-60	SU	1950	57 mm	Battery	3	12	T	8/5/3	M	18	5	10	15		2	2		1	5	—	—	—
ZSU-57-2	SU	1955	57 mm × 2	Section	4!	12	U	6/3	M	18	5	10	15		2	1		1	5	—	—	—
M51	US	1953	75 mm	Battery	3	18	T	9/6/3	M	30	6	12	18		4	4		3	6	—	—	—

AAA Units Table

	Type	Country	Year	Gun Size[α]	Defense Strength	Sighting Range	Mobility	VPs 3D/2D/D	AAA Class	Altitude	Range S	Range M	Range L	Hit S	Hit M	Hit L	Damage Rating	FCR	Night IR Sights	SAM				
KS-12	85 mm	SU	1939	Battery	4	18	T	9/6/3	H	27	6	12	18				2	1		0	6	—	—	—
90 mm M2	90 mm	US	1943	Battery	4	18	T	9/6/3	H	43	7	14	21				3	2		1	6	—	—	—
QF 3.7 Inch	94 mm	UK	1939	Battery	4	18	T	9/6/3	H	45	9	18	27				3	2		1	6	—	—	—
KS-19	100 mm	SU	1948	Battery	4	18	T	10/7/3	H	39	7	14	21				1	1		0	6	—	—	—
120 mm M1	120 mm	US	1944	Battery	4	18	T	9/6/3	H	57	11	22	33				2	2		1	7	—	—	—
KS-30	130 mm	SU	1955	Battery	4	18	T	11/7/4	H	64	12	24	36				1	1		0	7	—	—	—

FCR UNITS

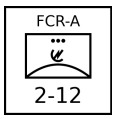
Most medium and heavy AAA batteries can be associated with an add-on fire-control radar (FCR), which improves their accuracy and allows them to engage unsighted targets at night and in adverse weather. These units are presented below with their counters.

The FCR Unit Table summarizes the properties of FCR units. As well as the basic properties common to all ground units, the table gives: the FCR class; the frequency band; and the modifier to the hit roll.

FCR Unit Table

Type	Size	Defense Strength	Sighting Range	Mobility	VPs 3D/2D/D	FCR Class	FCR Frequency	FCR Modifier
Towed FCR-A	Platoon	2	12	T	6/4/2	A	LF	−1
Towed FCR-B	Platoon	2	12	T	6/4/2	B	MF	−2
Towed FCR-C	Platoon	2	12	T	7/5/2	C	VF	−2
Towed FCR-D	Platoon	2	12	T	8/5/3	D	MW	−3

FCR-A

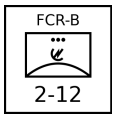


Towed FCR-A Platoon: FCR-As typically use 1940s technology. They include the SCR-584 and SON-9 FCRs.

SCR-584: The SCR-584 is an early US towed FCR-A based on the British 3 GHz magnetron. It replaced the much larger and less accurate SCR-268. It is commonly used to control batteries of 90 mm M2, QF 3.7 inch, and 120 mm M1 guns. The SCR-584 entered service with US and British anti-aircraft artillery units in 1944.[67]

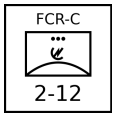
SON-9: The Soviet SON-9 (Fire Can) FCR-A was derived from the earlier SON-4, which in turn was derived from the US-supplied SCR-584. It is commonly used with 57 mm S-60, 85 mm KS-12, 100 mm KS-19, and 130 mm KS-30 batteries. The SON-9 entered service in 1955 with Soviet forces and was widely exported to Soviet allies. It has seen combat, including with North Vietnam in the Vietnam War, Egypt and Syria in the 1967 War, the War of Attrition, and the 1973 War.[68]

FCR-B



Towed FCR-B Platoon: FCR-Bs typically use 1950s technology.

FCR-C



Towed FCR-C Platoon: FCR-Cs typically use 1960s technology. They include the Super Fledermaus and Flycatcher FCRs.

Super Fledermaus: The Contraves Super Fledermaus FCR-C can control a section of two Oerlikon GDF 35 mm guns; it cannot control a whole battery. It entered service in 1965 with the Swiss Air Force and also was used by the Argentine Air Force, Brazil, Iran, Israel, Japan, and South Africa. It was used by Israel in the 1967 Six Day War and afterward with the Bofors 40 mm L/70 gun to defend strategic sites. It also saw combat with the Argentine Air Force in the 1982 South Atlantic War controlling Oerlikon GDF guns in the defense of BAM Malvinas (Port Stanley Airport).[69]

Flycatcher: The Signaal Flycatcher FCR-C can control up to six Bofors L/60 or L/70 guns or SAM launchers. Flycatcher entered service with the Royal Netherlands Air Force in 1979, serving for airfield defense and each controlling three modified Bofors 40 mm L/70 gun mounts. It subsequently served with the Indian Armed Forces from 1985, the Irish Defense Forces from 2003, the Royal Netherlands Army from 1987, the Thai Armed Forces from 1981, Turkey, and Venezuela.[70]

FCR-D

Towed FCR-D Platoon: FCR-Ds typically use 1970s technology. They include the Skyguard.

Skyguard: The Contraves Skyguard FCR-D can control two Oerlikon GDF 35 mm guns and additionally a SAM launcher with AIM-9 Sparrow, RIM-9 Sea Sparrow, or Aspide missiles. Skyguard entered service in 1977 with the Swiss Air Force. It was subsequently very widely exported and used by the Argentine Army, Austria, Bahrain, Bangladesh, Brazil, Canada, Chile, China, Cyprus, Egypt, Greece, Indonesia, Iran, Italy, South Korea, Kuwait, Malaysia, Oman, Pakistan, Saudi Arabia, Spain, Taiwan, and the UK. It saw combat with the Argentine Army in the 1982 South Atlantic War in the defense of BAM Malvinas (Port Stanley Airport) and BAM Condor (Goose Green Airfield) controlling Oerlikon GDF 35 mm guns.[71]

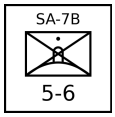
INFANTRY SAM UNITS

Infantry SAM units are squads of infantry specialized in the destruction of attacking aircraft with portable SAMs. These units are described below with their counters.

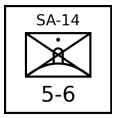
The Infantry SAM Unit Table summarizes the properties of infantry SAM units. In addition to the basic properties shared by all ground units, it gives: the number of ready, volley, and reload missiles; the multi-target capability; whether the unit has quick-reaction capability; whether the unit has IFF capability; whether the unit has night IR sights; and the optical lock-on number and range. Each squad has one launcher and, unless otherwise specified, one ready missile and two reload missiles. Infantry SAM squads may be attached to infantry, infantry weapons, infantry mortar, or infantry HQ platoons.

The Infantry SAM Table summarizes the properties of the actual missiles: guidance mode; launch roll; turn rate; flight time; visibility; ECCM, chaff, and flare ratings; whether the missile is active homing (AH) or has a home-on-jam (HOJ) capability; boost phase in FPs; base speed and sustainer duration in game turns; minimum altitude (and modifier for targets at T level); hit rolls for direct and proximity hits; and damage ratings for direct and proximity hits. Most infantry SAMs do not have true proximity fuzes, and so proximity hits in the game correspond to glancing direct hits in reality.

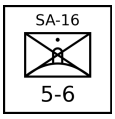
SA-7A Squad: The 9K32 Strela-2 (SA-7A Grail) was the first Soviet portable SAM. It has a rear-aspect, uncooled IR seeker. It entered service with Soviet forces in 1968, but quickly was replaced by the improved Strela-2M (SA-7B).[72]



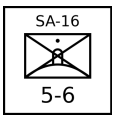
SA-7B Squad: The 9K32M Strela-2M (SA-7B Grail) is similar to the Strela-2 (SA-7A Grail), but uses a rear-aspect, electrically-cooled IR seeker. In Warsaw Pact armies, the Strela-2M was sometimes used with the PRP portable RWR receiver that was capable of passively identifying enemy aircraft by their radar emissions. The Strela-2M entered service with Soviet forces in 1970, replacing the Strela-2. It has seen combat in almost every conflict since 1970 in which the Soviet Union or its successor states have been participants or supporters. In particular, the Strela-2M was used heavily by North Vietnamese forces in the 1973 Easter Offensive, by Arab forces in the 1973 Yom Kippur War, by Angolan forces in the South African Border War, by the Afghan Mujahedeen in the Soviet-Afghan War (having been supplied by the CIA), by Iran in the Iran-Iraq War in the 1980s, and by Serbian and Serbian-supported forces in the Yugoslav Wars in the 1990s.[73]



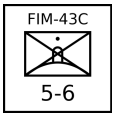
SA-14 Squad: The 9K34 Strela-3 (SA-14 Gremlin) is a development of the earlier Strela-2M. It has a rear-aspect, cryogen-cooled IR seeker. In Warsaw Pact armies, the Strela-3 was sometimes used with the PRP portable RWR receiver that was capable of passively identifying enemy aircraft by their radar emissions. The Strela-3 entered service with the Soviet Army in 1974. It has seen combat in many conflicts since then in which the Soviet Union or its successor states have been participants or supporters. In particular, it was used by Iraqi forces in the 1991 Gulf War.[74]



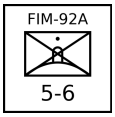
SA-16 Squad: The 9K310 Igla-1 (SA-16 Gimlet) uses as single-channel IR seeker similar to that in the SA-14 in a new missile body. The warhead is similar to the SA-7/14, but also exploded any remaining rocket fuel. In the game, this is simulated by increasing the damage rating by one if the missile attacks before expending more than half of its FPs (round down). It incorporates an IFF interrogator, but this is only used in Warsaw Pact armies. The Igla-1 entered service with the Soviet Army in 1981. It has been widely exported and have seen combat in many conflicts since then in which the Soviet Union or its successor states have been participants or supporters. [75]

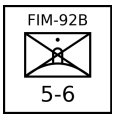


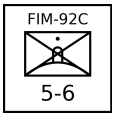
SA-18 Squad: The 9K38 Igla (SA-18 Grouse) uses a similar missile and launcher similar to the 9K310 Igla-1 (SA-16), but with a new a dual-channel IR seeker to reduce its susceptibility to flares. Again, the warhead is similar to the SA-7/14, but also exploded any remaining rocket fuel. In the game, this is simulated by increasing the damage rating by one if the missile attacks before expending more than half of its FPs (round down). It incorporates an IFF interrogator, but this is only used in Warsaw Pact armies. The Igla entered service with the Soviet Army in 1983. It has been widely exported and have seen combat in many conflicts since then in which the Soviet Union or its successor states have been participants or supporters. It was used by Iraqi forces in the 1991 Gulf War, by the Peruvian Army in the 1995 Cenepa War, and by Serbian forces in the 1999 Kosovo War.[76]

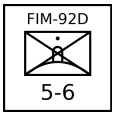


FIM-43C Squad: The FIM-43C Redeye Block III was the first US infantry SAM to reach production and enter service. It has a rear-aspect, cryogen-cooled IR seeker. The FIM-43C entered service with the US Army in 1968 and subsequently was used by Australia, Chad, Denmark, West Germany, Greece, Israel, Jordan, Saudi Arabia, Somalia, Sudan, Sweden, and Turkey. It was also supplied by the CIA to the Afghan Mujahedeen and the Nicaraguan Contras.[77]

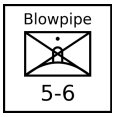






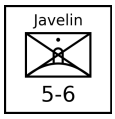


FIM-92A/B/C/D Squad: The FIM-92A Stinger has an all-angle, second-generation, cryogen-cooled IR seeker. Its warhead weight 3 kg and was significantly more lethal than in earlier missiles. The FIM-92B Stinger POST adds a UV seeker to distinguish flares from aircraft. The FIM-92C/D Stinger RMP have software updates to further improve the performance against flares. All have an IFF interrogator to help avoid fratricide. The FIM-92A entered service with the US Army in 1981, the FIM-92B in 1986, the FIM-92C in 1989, and the FIM-92D in 1992. They were exported to dozens of US allies including the Afghan Mujahedeen and the Angolan UNITA. They saw service in the 1982 South Atlantic War, the Soviet-Afghan War, the Angolan Civil War, the Libyan Invasion of Chad, the Tajik Civil War, the Chechen War, the Sri Lankan Civil War, the Syrian Civil War, and the Russian Invasion of Ukraine.[78]

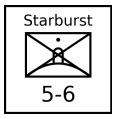


Blowpipe Squad: Blowpipe is an MCLOS optically-guided missile. While in theory this gave it an all-aspect capability, in practice guiding the missile to the target in combat proved to be very difficult. Blow-

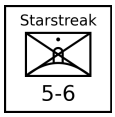
pipe entered service with the British Army and Royal Marines in 1975. It was also used by the Argentine Army, Canadian Army, Chilean Army and Navy, Ecuadorean Army, Guatemalan Army, Portuguese Army, Malawian Army, Malaysian Army, Nigerian Army, Omani Army, Royal Thai Air Force and Army, and the UAE Army. It saw combat in the 1982 South Atlantic War with both sides and had a low success rate. In 1986, it was trialed by the Afghan Mujahedeen, but again was not considered to be effective. Finally, it was also used in the 1995 Cenepa War by Ecuador.[79]



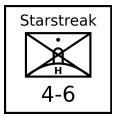
Javelin Squad: Given the poor performance of Blowpipe in the 1982 South Atlantic War, Javelin was quickly developed as an upgraded version of Blowpipe with SACLOS guidance in place of MCLOS guidance. Javelin replaced Blowpipe in the British Army and Royal Marines in 1984. It was also used by the Botswana Army, the Canadian Army, the Malaysian Army, the Peruvian Army, and the South Korean Army.[80]



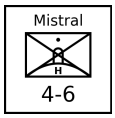
Starburst Squad: Starburst is Javelin with the radio-link replaced by a with laser link. Starburst replaced Javelin in the British Army and Royal Marines in 1989. It was also used by the Canadian Army, the Malaysian Army, the Qatari Army, and the Royal Thai Army. Starburst served with the British Army in the 1991 Gulf War, but was not used in combat.[81]



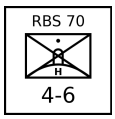
Starstreak Squad: Starstreak is a high-speed laser-guided missile. As the missile approaches the target, it launches three independent explosive darts. The darts have contact fuzes, but not proximity fuzes. Starstreak replaced Javelin in the British Army and Royal Marines in 2000 and is also used by the South African Army and the Armed Forces of Ukraine. Starstreak served with the British Army in the 2003 Invasion of Iraq, but was not used in combat. It has seen combat in the Russian Invasion of Ukraine.[82]



Starstreak LML Squad: The Starstreak Light Multiple Launcher (LML) has three ready-to-fire Starstreak missiles on a pedestal mount with a single sighting unit. The launcher and its associated equipment are heavy and will normally be transported by a vehicle unit. Starstreak LML entered service with the British Army and Royal Marines in 2000. It is also used by Indonesia, Malaysia, and South Africa.[83]



Mistral Squad: The Mistral is a French IR-homing missile on a pedestal mount. The missile has a considerably larger warhead than either the Stinger or Igla missiles, but this comes at a cost in portability. The launcher and its associated equipment are heavy and will normally be transported by a vehicle unit. The Mistral entered service with the French Army in 1990. It was subsequently exported to many other countries.[84]



RBS 70 Squad: The RBS 70 is a Swedish laser-guided missile on a pedestal mount. The launcher and its associated equipment are heavy and will normally be transported by a vehicle unit. The RBS 70 entered service with the Swedish Army in 1977. It was subsequently used by Argentina, Australia, Bahrain, Brazil, the Czech Republic, Iran, Ireland, Lithuania, Norway, Pakistan, Singapore, Thailand, Tunisia, the UAE, Ukraine, and Venezuela.[85]

Ayn-al-Saqr Squad: The Ayn-al-Saqr ("Eye of the Hawk") missile is a licensed version of the 9M32M Strela-2m (SA-7B Grail) manufactured in Egypt. It equips one version of the Sinai 23 air-defense vehicle. It entered service with the Egyptian Army in 1984.[86]

Infantry SAM Unit Table

Name	Size	Year	Defense Strength	Sighting Range	Mobility	VPs 3D/2D/D	SAM	Missiles Ready	Missiles Volley	Missiles Reload	Multi-Target	Quick Reaction	IFF	Night IR Sights	Lock-On Optical	Lock-On Range	Other Names
SA-7A	Squad	1968	5	6	U	–/–/3	SA-7A	1	1	2	—	Y	—	—	7	9	9K32 Strela-2 (Grail)
SA-7B	Squad	1972	5	6	U	–/–/4	SA-7B	1	1	2	—	Y	—	—	7	9	9K32M Strela-2M (Grail)
SA-14	Squad	1974	5	6	U	–/–/6	SA-14	1	1	2	—	Y	—	—	7	12	9K34 Strela-3 (Gremlin)
SA-16	Squad	1981	5	6	U	–/–/8	SA-16	1	1	2	—	Y	Y	—	7	12	9K310 Igla-1 (Gimlet)
SA-18	Squad	1983	5	6	U	–/–/10	SA-18	1	1	2	—	Y	Y	—	8	16	9K38 Igla (Grouse)
FIM-43C	Squad	1968	5	6	U	–/–/4	FIM-43C	1	1	2	—	Y	—	—	7	9	Redeye Block III
FIM-92A	Squad	1981	5	6	U	–/–/9	FIM-92A	1	1	2	—	Y	Y	—	8	12	Stinger
FIM-92B	Squad	1986	5	6	U	–/–/10	FIM-92B	1	1	2	—	Y	Y	—	8	12	Stinger POST
FIM-92C	Squad	1989	5	6	U	–/–/10	FIM-92C	1	1	2	—	Y	Y	—	8	12	Stinger RMP
FIM-92D	Squad	1992	5	6	U	–/–/10	FIM-92D	1	1	2	—	Y	Y	—	8	12	Stinger RMP
Blowpipe	Squad	1975	5	6	U	–/–/6	Blowpipe	1	1	2	—	Y	Y	—	7	9	
Javelin	Squad	1984	5	6	U	–/–/8	Javelin	1	1	2	—	Y	Y	—	7	9	Javelin GL
Starburst	Squad	1989	5	6	U	–/–/8	Starburst	1	1	2	—	Y	Y	—	7	9	Javelin S15
Starstreak	Squad	2000	5	6	U	–/–/12	Starstreak	1	1	2	—	Y	Y	—	7	12	
Starstreak LML	Squad	2000	4	6	U	–/–/14	Starstreak	3	1	0	—	Y	Y	—	7	12	
Mistral	Squad	1990	4	6	U	–/–/11	Mistral	1	1	2	—	Y	—	—	7	12	
RBS 70	Squad	1977	4	6	U	–/–/8	RBS 70	1	1	2	—	Y	Y	—	7	10	
Ayn-al-Saqr	Squad	1984	5	6	U	–/–/4	Ayn-al-Saqr	1	1	2	—	Y	—	—	7	12	

Infantry SAM Table

Name	Year	Guidance Mode	Launch Roll	Turn Rate	Flight Time	Visibility	ECCM	Chaff	Flare	AH	HOJ	Instant Arming	Boost Phase	Base Speed	Sustainer	Minimum Altitude	Hit Direct	Hit Proximity	Damage Direct	Damage Proximity	Other Names
SA-7A	1968	I	6	BT	1	6	—	—	5	—	—	Y	—	10	0	T	5	6	4	2	9M32 Strela-2 (Grail)
SA-7B	1972	I	7	BT	1	6	—	—	5	—	—	Y	—	10	0	T	5	6	4	2	9M32M Strela-2M (Grail)
SA-14	1974	M	7	BT	1	6	—	—	4	—	—	Y	—	9	0	T	6	7	4	2	9M34 Strela-3 (Gremlin)
SA-16	1981	M	8	BT/2	1	6	—	—	4	—	—	Y	—	12	0	T	6	7	4*	2*	9K310 Igla-1 (Gimlet)
SA-18	1983	A	8	BT/2	1	6	—	—	2	—	—	Y	—	12	0	T	7	8	4*	2*	9K38 Igla (Grouse)
FIM-43C	1968	M	7	BT	1	6	—	—	5	—	—	Y	—	10	0	T	6	7	4	2	Redeye Block III
FIM-92A	1981	A	8	BT/2	1	2	—	—	3	—	—	Y	—	14	0	T	7	8	5	3	Stinger
FIM-92B	1986	A	8	BT/2	1	2	—	—	1	—	—	Y	—	14	0	T	7	8	5	3	Stinger POST
FIM-92C	1989	A	8	BT/2	1	2	—	—	1	—	—	Y	—	14	0	T	7	8	5	3	Stinger RMP
FIM-92D	1992	A	8	BT/2	1	2	—	—	1	—	—	Y	—	14	0	T	7	8	5	3	Stinger RMP
Blowpipe	1975	OG	8	HT	1	6	—	3	0	—	—	—	—	8	0	T	3	7	6	5	
Javelin	1984	OG	8	ET	1	5	—	2	0	—	—	Y	—	10	0	T	5	8	6	5	Javelin GL
Starburst	1989	LG	8	ET	1	5	—	1	0	—	—	Y	—	10	0	T	5	8	6	5	Javelin S15
Starstreak	2000	LG	8	ET	1	5	—	1	0	—	—	Y	—	24	0	T	7	9	6	4	
Mistral	1990	A	8	ET	1	5	—	—	2	—	—	Y	—	18	0	T	7	9	6	3	
RBS 70	1977	LG	8	ET	1	6	—	2	0	—	—	—	—	12	0	T	5	8	6	3	
Ayn-al-Saqr	1984	I	7	BT	1	6	—	—	5	—	—	Y	—	10	0	T	5	7	4	3	

EXAMPLE ORDERS OF BATTLE

This appendix contains a number of example orders of battle and their correspondence in game units.

The orders of battle for battalions are presented with the headquarters unit in the top row, any battalion-level assets on the next row or rows, and finally the line companies on the remaining rows. Within each row, infantry or towed units appear on the left and their transport units on the right.

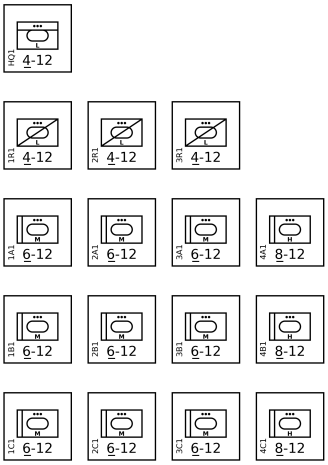
The unit identifiers for line platoons are the platoon number, followed by the company letter or number, followed by the battalion number. If only one battalion is used, the battalion number may be omitted. Transport platoons will have a "T" prefix. For example, "1B3" means the first platoon of B company of the third battalion, and "T1B3" is its transport platoon. Companies normally have letters in Western armies and numbers in Warsaw Pact armies. Additionally, AD is used for air defense, AR for artillery, AT for anti-tank, R for reconnaissance, and S for scout, support, or security. The unit identifiers for battalion HQ units are HQ followed by the battalion number.

Within the game, units are referred to as a squad, section, platoon/battery, or battalion according to their size. These terms are not used uniformly by all armies. For example, in armored formations in the British Army, the terms troop, squadron, and regiment are used in place of platoon, company, and battalion. Furthermore, even when these terms are used, they do not always refer to units of the same size. To help navigate these differences, when a different term is used for a formation, the game equivalent will appear in brackets.

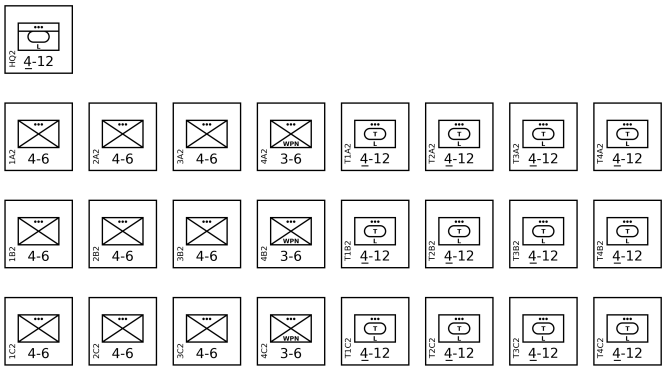
BRITISH ARMY (EARLY 1960S)

The British Army in the early 1960s was in transition. It still operated with a mixture of medium Centurion, upgraded with the L7 105 mm gun, and heavy Conqueror tanks, and operated both mechanized and motorized infantry.[87]

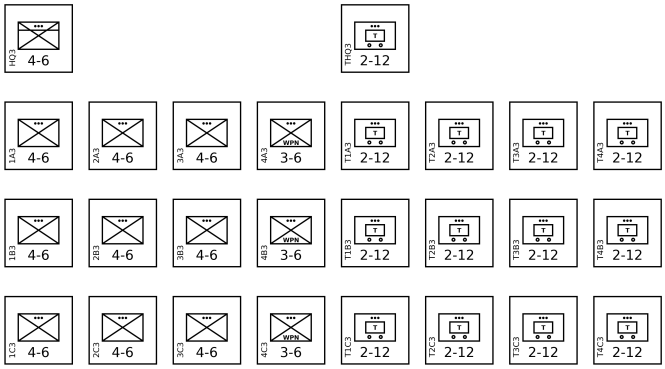
Armoured Regiment: This regiment [battalion] has three squadrons [companies] A/B/C each consisting of three tank troops [platoons] with Centurion medium tanks and one with Conqueror heavy tanks. The reconnaissance (R) squadron [company] is equipped with Ferret armored cars.



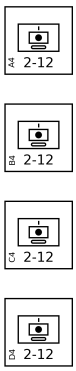
APC Infantry Battalion: This battalion has three companies A/B/C each consisting of three infantry platoons and one infantry weapons platoon, with Saracen light APCs to provide mobility and protection.



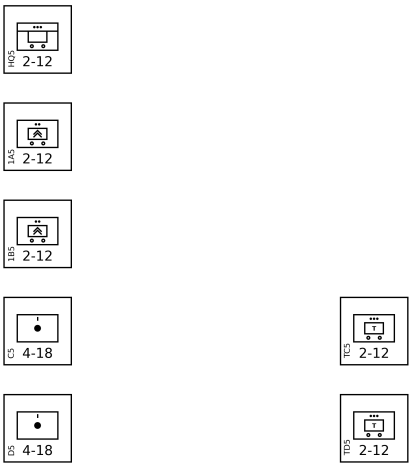
Infantry Battalion: This battalion has three companies A/B/C each consisting of three infantry platoons and one infantry weapons platoon, with trucks to provide mobility.



Field Artillery Regiment: This regiment [battalion] has four batteries A/B/C/D each with six Cardinal (M44) self-propelled guns.



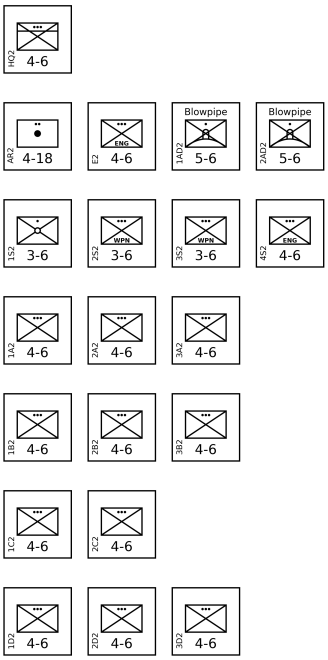
Heavy Artillery Regiment: This regiment [battalion] has four batteries A/B/C/D, of which A/B batteries [sections] each have two Honest John launchers and C/D batteries each have four towed 8-inch M1 guns with trucks.



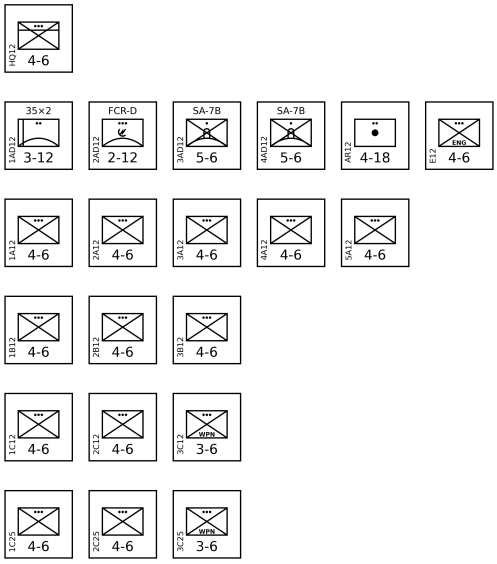
BATTLE OF GOOSE GREEN

The 1982 Battle of Goose Green in the South Atlantic War was fought between 2 Para Battalion of the British Army and 12 Infantry Regiment of the Argentine Army, both with attached elements.[88]

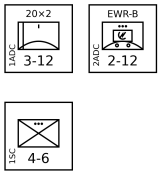
2 Para Battalion: 2 Para at the time consisted of three line companies A/B/D each with three platoons of infantry, one patrol company C with two infantry platoons, and a support company S with one platoon of mortars [but with only 2 out of 8 mortars available, so equivalent to a squad], one platoon of sustained-fire GPMGs, one platoon with MILAN portable ATGMs, and one engineers platoon. The attached units were half a battery [equivalent to a section] of three 105 mm Light Guns from Alma Battery of 29 Commando Regiment, Royal Artillery, two squads of Blowpipes from 43 Air Defence Battery, Royal Artillery, and a Recce Troop [platoon] of 59 Independent Commando Squadron, Royal Engineers.



Task Force Mercedes: The Argentine Army at Goose Green was called *Fuerza de Tareas Mercédes* (Task Force Mercedes). It consisted of *Regimento de Infantería 12* (12th Infantry Regiment), with three line companies A/B/C, and reinforced with C company of 25th Infantry Regiment, three 105 mm pack howitzers [one section] of A Battery of 4th Airborne Artillery Regiment, an engineers platoon from the 9th Engineering Company, two Oerlikon GDF 35 mm AA guns with a Skyguard FCR of B Battery of 601st Anti-Aircraft Artillery Group, and a pair of SA-7B squads.



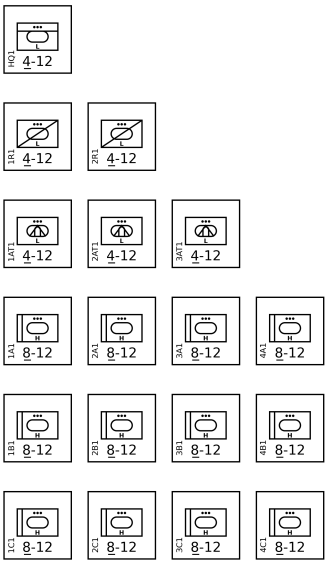
Military Airbase Condor: The Argentine Air Force renamed the airstrip at Goose Green as *Base Aérea Militar Cóndor* (Condor Military Air Base). The air-force elements at the base included a battery of two 20 mm Rh-202 AA guns of 3 Section B Battery 601AAAG with an Elta EWR and a security platoon.



BRITISH ARMY (MID 1980S)

The British Army in the mid 1980s.[89]

Armoured Regiment: This regiment [battalion] has three squadrons [companies] A/B/C each consisting of four tank troops [platoons] each with fourteen Chieftain heavy tanks. The reconnaissance (R) squadron [company] has two troops [platoons] each equipped with four Scorpion light tanks. The anti-tank squadron [company] has tree troops [platoons] each equipped with three FV438s with the Swingfire missile.


HQ1 4-12
1R1 4-12
2R1 4-12
1AT1 4-12
2AT1 4-12
3AT1 4-12
1A1 8-12
2A1 8-12
3A1 8-12
4A1 8-12
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3C1 8-12
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ADDITIONAL RULES

RULE 1: GROUND UNITS

A ground *unit* is a combat or logistics formation of infantry, armor, transports, AAA, SAM launchers, or radars. This section discusses them in general.[90]

A. Type and Size: A ground unit will have a type, for example, "heavy tank" or "ZSU-23-4", and a size, which will be one of squad, section, platoon, or battery. Typically, a squad has five to ten soldiers and/or one vehicle, gun, or launcher; a section has ten to twenty soldiers and/or two vehicles, guns, or launchers; and a platoon or battery has thirty to fifty soldiers and/or three to six vehicles, guns, or launchers. The term "platoon" is more commonly used for infantry and tank units and "battery" for artillery and air-defense units, but within the game they are equivalent.

Companies, battalions, and other larger units above platoons or batteries are represented by multiple units. For example, an infantry company might be represented by three infantry platoon units.

All ground-combat and transport platoons are also available as sections and squads and all AAA platoons and batteries are also available as sections. A section or squad is identical in every way to a platoon or battery that has taken 1D or 2D damage, respectively, except that no VPs are awarded for this notional damage and non-AAA squads and sections may not engage in barrage fire.[91]

In the game, ground units are uniformly referred to as squads, sections, platoons, and batteries. However, other terms are used for these units even within English-speaking armies. For example, in many Commonwealth armies a section is equivalent to a squad, a troop is equivalent to a platoon, and a squadron is equivalent to a company. In the US Armored Cavalry, a troop is equivalent to a company and a squadron is equivalent to a battalion.

B. Basic Properties: The type and size of a ground unit determines its basic properties: damage resilience, defense strength, target class, mobility, and sighting range.

The damage resilience is the amount of damage a ground unit can suffer before being killed. It is determined by the size of the unit: a squad has a damage resilience of 1D; a section has a damage resilience of 2D; and a platoon or battery has a damage resilience of 3D.[92]

The numerical defense strength is used to calculate the odds for air-to-ground attacks and for ground-to-ground combat.

The target class is used to determine the appropriate attack strength of any weapons used in air-to-ground attacks. The class can be a hard, open, or soft according to the degree to which the unit enjoys armored protection. Many weapons have different attack strengths against targets of different classes. A hard target has armored protection from both direct (front, side, and rear) and top attacks. Hard targets are further described as light, medium, or heavy according to their degree of protection, and have corresponding defense strengths.[93] An open target has armored protection from direct attacks but not from top attacks. An open target is normally treated as a hard target, but against napalm, fire, FAE, and air-burst HE bombs, it is considered to be a soft target.[94] A soft target does not have armored protection their personnel and/or weapons.[95]

C. Representation: The briefing documents show the graphical representation of ground units as counters.[96] The size of a unit is indicated by one, two, or three dots (for squads, sections, or platoons) or one vertical bar (for batteries) in an upper position. Hard targets are indicated by an underlined defense strength and open targets by an underline defense strength followed by an exclamation point.

Advanced Rule

D. Mobility: The mobility of a ground unit restricts the hexes it can occupy according to the terrain in and around those hexes. Infantry units, tracked armored units, wheeled armored units other than armored trucks, and tracked unarmored units have unrestricted (U) mobility. Trucks, armored trucks, and other wheeled unarmored units have good (G) or poor (P) mobility. All other units have towed (T) mobility.[97]

A unit with unrestricted mobility may occupy: any urban hex; any clear hex; or any forest hex. A unit with good mobility may occupy: any urban hex; any clear hex; or any forest hex with a road or trail. A unit with poor mobility may occupy: any urban hex; any clear or forest hex with a road, trail, or runway; or any clear hex adjacent to an urban hex or another clear hex with a road, trail, or runway. A unit with towed mobility may occupy hexes according to the mobility of its transport.[98]

When necessary, a scenario must specify whether trucks, armored trucks, and other wheeled unarmored units have good or poor mobility, according to the terrain and the capabilities of the vehicles represented, and assign transport units to towed units.[99] A scenario may modify the mobility classes of ground units and the associated restrictions. [100]

E. Composite Platoons: At the start of the game, an infantry, infantry weapons, infantry mortar, or infantry HQ platoon may attach one or two infantry squads to form a composite platoon. The attached squads are commonly infantry SAM or FAC squads. Once formed, a composite platoon may not be split into its constituent units.[101]

An attached squad is not represented by a separate counter, does not count for stacking, and may not be sighted, identified, or attacked directly. Instead, it is considered to form part of the composite platoon. An attached squad retains its normal capabilities (for example, an attached SAM squad may engage aircraft and an attached FAC squad may sight and mark targets).

A composite platoon has the same damage resilience, defense strength, target class, mobility, and sighting range as its constituent platoon. If a composite platoon is sighted, no information is given on any attached squads. If a composite platoon is identified, any attached squads are also identified (for example, "infantry platoon with an attached FAC squad"). If a composite platoon is suppressed, each attached squad is suppressed too. If a composite platoon suffers damage, then for each D of damage, roll one die for each attached squad and on a 3– the attached squad is killed. If a composite platoon is killed, each attached squad is killed too. VPs are awarded for damage to each of the constituent units.[102]

F. Transport: An infantry or towed unit may be transported by a specific associated transporting IFV, APC, armored truck, truck, or tracked-carrier unit of at least the same size. When necessary,

a scenario will establish the association between units and their specific transports. [103]

A transported unit is not represented by a separate counter, does not count for stacking, and may not be sighted, identified, or attacked directly. It may not attack other ground units or aircraft, may not launch SAMs, may not use radar, and may not act as a FAC.

If a transporting unit is sighted, no information is given on the transported unit. If a transporting unit is identified, no information is given on a transported infantry unit, but any transported towed unit is also identified (for example, "truck platoon towing an artillery battery" or "truck platoon towing a ZPU-4 battery"). If a transporting unit suffers damage, then its transported unit suffers the same damage. VPs are awarded for damage to both the transporting and transported unit.

A unit may start the game either transported or not. If not transported, it may be placed according to either its mobility or that of its associated transport.[104]

An infantry unit in the same hex as its associated transport may declare that it is mounting the transport at the start of a Ground Unit Interaction Phase. Both units must not be suppressed, and the transport must not have suffered more damage than the infantry unit. At the end of the Ground Unit Interaction Phase five game turns later, the unit is considered to be transported, and its counter is removed from the map.

An infantry unit being transported may declare that it is dismounting at the start of a Ground Unit Interaction Phase. The transport unit must not be suppressed. At the end of the Ground Unit Interaction Phase two game turns later, the unit is considered to be no longer transported, and its counter is placed on the map in the same hex as its transport. If placing the counter would violate stacking requirements, the unit must abort the dismounting process.

During these processes, the infantry unit may not perform any other action and the transport may not move and may not carry out ground attacks, but may perform barrage fire if it has this capability. The infantry unit may abort either of these process at the start of any Ground Unit Interaction Phase. If either are suppressed during the process, the unit must abort the process in the next Ground Unit Interaction Phase.[105]

A towed unit may not mount or dismount during the game.

NOTES

1. **Robin Olds**

The full quotation contains a second insight: "You're not going to win a war by shooting down a damn MiG, you're going to win a war by bombing the bejesus out of whatever you were sent to bomb. Of course, we weren't there to win it anyway, so it was all a waste of time." (Sherwood, John. Fast Movers: Jet Pilots and the Vietnam Experience, chapter 1)

2. **Ground-Combat and Transport Units**

Air Strike and *Eagles of the Gulf* provide eleven types of ground-combat and transport unit: light, medium, and heavy armor, infantry, infantry FAC, infantry SAM, towed artillery, infantry HQ, armored HQ, truck, and light truck. We provide many more. Does this change outcomes? No. Effectively, all ground units are basically a protection class, defense strength, and sighting range, and the new units map to the existing ones. Does this change the experience? I think it does; the wider range of units enhances the gaming experience.

3. **Infantry Weapons Platoon**

An infantry weapons platoon is not strictly necessary, but it adds to the look-and-feel, and we might consider giving them longer range in ground-to-ground combat. Having a single type is probably more appropriate than having separate types for grenade launchers, recoilless rifles, ATGMs, and machine guns. One might think that the .50 cal or DShK infantry AAA unit could serve as a machine-gun weapons platoon, but machine-gun weapons platoons have low tripods for ground combat rather than high ones for AAA.

4. **Infantry HQs**

In the *Air Strike* scenarios HQ platoons are included at a ratio of about one to every nine normal platoons, so they seem to be battalion HQs.

5. **Tanks**

Protection Levels:

Light, medium, and heavy armor correspond to defense strengths of 4, 6, and 8. Considering the vehicles depicted in scenarios in *Air Strike* and *Eagles of the Gulf*, it seems that typically:

- Light = protection from 14.5 mm and smaller rounds.
- Medium = protection from autocannons and guns up to about 90 mm.
- Heavy = protection against guns of 100 mm or more.

Obviously, this is a gross simplification as each vehicle has its own strengths and weaknesses, and its armor varies between its front, side, rear, and top.

To extend this to other tanks and AFVs, we use Wargames Research Group (WRG) 1950–1985 miniatures rules, which give armor classes of I to X for a range of AFVs:

CLASS I
Sexton
Archer
VP.90
AMX.105A
AMX.155F3
BTR.40
BTR.152V
BTR.50P
BTR.60P
ASU.57
ZSU.57-2
SU.76
AT-P
M.7
M.42
M.44
M.56
M.107
M.110
Ontos
XR.311
½ tracks
RBY.1

CLASS II
Ferret
Fox
Saracen
FV.432
Abbot
SP.70
FV.180
AMX.105B
AMX.155GCT
AMX.13DCA
AML
VAB
YP.408
Spz.kurz.
PVB.302
VK.155
BRDM
BMD
BMP.1
BTR.152K
BTR.50PK
BTR.60PB
PT.76
ZSU.23-4
SP.73
SP.74
Commando
M.8
M.18
M.52
M.55
M.59
M.108
M.109
M.113
M.114
LVTP.7
Lynx
Cougar
Grizzly
L.33

CLASS II "s"
Dutch AIFV

CLASS III
Scorpion
Scimitar
Striker
Spartan
Saladin
EBR
AMX.10P
AMX.10RC
AMX.13
AMX VC1
Luchs.
HS.30
IKV.91
Staghound
M.5
M.10
M.24
M.41
Sheridan

CLASS III "s"
FVS
XM.723
UK.MICV

CLASS IV
Cromwell
AMX.30
Marder
JPz.Rak.
Gepard
TAM
Pz.4H
ASU.85
Sherman

CLASS V
Vijayanta
Comet
JPz.Kan
Panzer 68
Type 61
T.34/85
SU.100

CLASS VI
Leopard 1
Type 74
ISU.122
ISU.152
M.46
M.47

CLASS VI "s"
Leopard.1A4

CLASS VI"S"
AMX.32

CLASS VII
Centurion 3
T.54
T.55
M.48
M.103

CLASS VIII
Centurion 13
T.62
M.60

CLASS IX
T.72
T.64
T.10
IS.3

CLASS IX "S"
Leopard 2
EPC
T.80

CLASS X
Chieftain
Conqueror
Shir.1
Strv.103

CLASS X "S"
UK.MBT.80
Shir.2
XM.1
Merkava
Leopard 3

The correspondence seems to be:

- Open = 4! = class I.
- Light = 4 = classes II to III.
- Medium = 6 = classes IV to VIII.
- Heavy = 8 = classes IX and X.

Since these rules are from 1979 and from unclassified sources, their reliability for the more modern vehicles is somewhat in doubt. Nevertheless, they are probably fine for older vehicles.

Heavy Tanks: The WRG 1950–1985 rules give class IX or X (i.e., heavy) protection to the Chieftain, Conqueror, EPC (Leclerc), IS-3, Leopard 2, Merkava 1, Shir 2 (Challenger 1), Stridsvagn 103, T-10, T-64, T-72, and T-80. Beyond these, we have:

- Al-Kalid/VT-1A and Type 99: These are based on the Type 90-II prototype, which in turn is based on the T-72 (heavy).
- Ariete: This weighs 54 tonnes, only slightly less than the M1A1 (heavy) which also has a crew of four, and therefore presumably has similar protection.
- Arjun: The weights 59–68 tonnes and has crew of four, similar to the M1A1 (heavy), and so presumably has similar protection.
- Challenger 2: This is derived from the Challenger 1 (heavy).
- IS-2: This has a similar weight to the IS-3 (heavy) and a similar armament (122 mm D-25), so presumably similar protection.
- K1: This is derived from the M1 (heavy).
- K2: This presumably has at least the same protection as the K1 (heavy).
- M-84, PT-91, and T-90: These are derived from the T-72 (heavy).
- Merkava 2/3/4: These are derived from the Merkava 1 (heavy).
- T-14: This presumably has at least the same protection as the T-90 (heavy).
- T-84: This is derived from the T-80 (heavy).
- Type 10: This weighs 44 tonnes in its standard configuration and has a crew of three, and so presumably has protection at least equal to earlier tanks with similar weights such as the T-72 (heavy).
- Type 90: The has a crew of three and weights 50 tonnes, and so presumably has protection at least the equal of the lighter T-72 (heavy).
- Type 99: The has a crew of three and weights 51–55 tonnes, and so presumably has protection at least the equal of the lighter T-72 (heavy).

Medium Tanks: The WRG 1950–1985 rules give class IV to IIX (i.e., medium) protection to the AMX-30, ASU-85, Centurion, Comet, Cromwell, ISU-122, ISU-152, Kanonen JPz, Leopard 1, M103, M4 Sherman, M46, M47, M48, M48, M60, Panzer 68, SU-100, T-34, T-54, T-55, T-62, TAM, Type 61, Type 74, and Vijayanta. Beyond these, we have:

- M26: The M46 (medium) is largely an M26 with an improved engine.
- M-50 and M-51 Sherman: These are based on the M4 (medium) and have improved armament and engines.
- Panzer 61: This is essentially an early version of the Panzer 68 (medium).
- Type 59/69/79: The Type 59 was Chinese version of the T-54A (medium). The Type 69 and 79 had improved equipment, some of which was derived from captured T-62s, but the same level of protection.
- Type 85/88/96: These have new hulls and turrets. It's not clear whether their level of protection is significantly improved over the Type 59/69/79 (medium).
- Type 15: Wikipedia gives little information on the armor, but the weight is 36 tonnes (with the armor package), and that seems to correspond more to a medium tank.
- Vickers MBT: This is similar to the Vijayanta (medium).

Light Tanks: The WRG 1950–1985 rules give class II and III (i.e., light) protection to the AMX 10RC, AMX 13, Commando, Cougar, Fox, SpPz Luchs, M8, M24, M41, M114A2, M551 Panhard AML, Panhard EBR, PT-76, Saracen, Scimitar, and Scorpion Beyond these, we have:

- Stryker MGS: With additional ceramic armor, this has protection against 14.5 mm but not heavier rounds, which I believe counts as III.
- Strv 74: This has a similar weight to the M41 (light).
- Type 62: Wikipedia give the hull armor as being similar to the M41 (light).
- Type 63: Wikipedia give the hull armor as being only 10 mm, which corresponds to light.
- Type 05: Based on Type 05 IFV (light).
- Type 08: Based on Type 08 IFV (light).

Open Tanks: Open tanks are uncommon, but we need an open target class for M3 half-track APCs, so it is easy to apply it here too. The WRG 1950–1985 rules give class I (i.e., open) protection to the ASU-57 and SU-76.

6. **IFV Units**

We might distinguish APCs and IFVs in ground-to-ground combat. Bradley IFVs made mincemeat of Iraqi tank regiments; M113 APCs would not have had the same impact.

Medium IFVs: The WRG 1950–1985 rules give class IV (i.e., medium) protection to the Marder. Beyond these, we have:

- BMP-3: Wikipedia says the BMP-3 has frontal protection from 30 mm rounds at 200 m.
- Boxer: Wikipedia states that with the baseline armor it is protected against medium-caliber rounds. Additional armor is also available.
- Strf 9040/CV90: Wikipedia states that the CV90 has frontal protection against 30 mm APFSDS rounds. This is considerably more than typical light AFVs.
- LAV III (add-on-armor): Wikipedia says the basic version only has protection against small-arms rounds, but an additional armor kit developed in 2009 gives it protection against 30 mm rounds.
- M2A2/M2A3/M2A4: Wikipedia states their these M2A2 also has protection against 30 mm APDS rounds and RPGs. The M2A3 and M2A4 have visually similar armor.
- Puma: Wikipedia says the frontal arc has protection against medium-caliber rounds and shaped-charge projectiles. With additional armor, the sides gain similar protection.

Light IFVs: The WRG 1950–1985 rules give class II or III (i.e., light) protection to the AMX-10P, BMD, BMP-1, Commando, Dutch AIFV (YPR-765), FV432 (FV432 RARDEN), HS.30 (SPz Lang), M113 (M113 FSC), Spz Kurz, UK MICV (Warrior), and XM723 (M2/M3 Bradley). Beyond these, we have:

- BMP-2: Wikipedia says the BMP-2 has similar protection to the BMP-1.
- LAV-25: Wikipedia says this only has protection against small-arms rounds.
- LAV III: Wikipedia says the basic version only has protection against small-arms rounds, but an additional armor kit developed in 2009 gives it protection against 30 mm rounds.
- Pbv 301: Wikipedia states the armor ranges from 8 to 50 mm, with upper front armor being 34 mm (20 mm at 55 degree). This seems to be light.
- Piranha IFV: Presumably similar protection to LAVs.
- Strkyer ICVD: This is derived from the Stryker (light).
- Type 86: Wikipedia states this is reverse-engineered from the BMP-1.
- Type 92: Wikipedia states that it is protected against small-arms rounds and the frontal armor can stop 12.7 mm rounds at 100 meters and 25 mm rounds at 1000 meters. This seems to correspond to light armor. The weight is also 21 tonnes, which is considerable less than most medium IFVs.
- Type 05: Wikipedia states that it has protection against small-arms rounds and some protection against 12.7 mm rounds. This corresponds to light armor.
- Type 08: Wikipedia states that it is protected against small-arms rounds and the frontal armor can stop 12.7 mm rounds at 200 meter. This corresponds to light armor.
- VCBI: Wikipedia says this only has protection against 14.5 mm rounds.

There are many variants of the Piranha, and the in-service date is a guess.

7. **APC Units**

Heavy APCs: These are converted tanks.

- Achzarit: Wikipedia states that this based on the T-54/T-55 (medium), but has additional composite armor. Its weight at 44 tonnes is much heaver than the T-55 at 36 tonnes, despite the lack of turret, which suggests the additional armor is quite significant. Therefore, we consider it to be heavy.
- Namer: This is based on the Merkava IV (heavy).
- Nakpadon: This is an improved version of the Nagmashot and Nagmachon. Wikipedia states that its weight is around 55 tonnes, which strongly suggests its armor qualifies as heavy.
- Puma: The photos in Wikipedia show additional armor. Its weight at 50 tonnes is almost identical to the Centurion, despite the lack of turret, which suggests the additional armor is quite significant.

The in-service dates for the Nakpadon (1998) and Puma (1993) are guesses.

Medium APCs:

- Boxer: Wikipedia states that with the baseline armor it is protected against medium-caliber rounds. Additional armor is also available.
- Nagmashot: Based on the Centurion (medium).
- Nagmachon: Based on the Centurion (medium).

The in-service dates for the Nagmashot (1990) and Nagmachon (2000) are guesses.

Light APCs: The WRG 1950–1985 rules give class II or III (i.e., light) protection to the BTR-50PK, BTR-60PB, Commando, Grizzly, FV432, LVTP-7, M113, M59, Saracen, Spartan, VAB, and YPR-408. Beyond these, we have:

- Bronco/Warthog: Wikipedia says it has protection against 7.62 mm rounds.
- BVS10/Viking: Wikipedia says it has protection against small-arms fire and can have additional armor to protect against 14.5 mm rounds.
- BTR-70: Wikipedia states the frontal armor is only 9 mm.
- BTR-80: Wikipedia states that this (and previous models) only have protection from small-arms fire.
- LAV II Bison: Wikipedia states that the Coyote is protected against small-arms rounds, and the Bison seems to be similar.
- LVTP-5: Wikipedia states that the maximum armor is 16 mm.
- M75: Wikipedia states the maximum armor is 38 mm. This is more than the thickness of some light tanks, but it is almost vertical.
- OT-64 SKOT: Wikipedia states the maximum armor is 13 mm.
- OT-810: Wikipedia states the maximum armor is 14.5 mm.
- Piranha APC: Presumably similar protection to LAVs.
- M1126 Stryker: Wikipedia states it has some protection against 14.5 mm rounds from the front and all-round protection against small-arms fire.
- TPz Fuchs: Wikipedia states that it is protected against small-arms rounds.
- Type 63: Wikipedia states that it is protected against small-arms rounds.
- Type 77: Wikipedia states that it is based on the PT-76 (light), and its weight of 15–5 tonnes matches other light APCs.
- Type 89: Wikipedia states that it is protected against small-arms rounds.
- Type 08: Based on Type 08 IFV (light).

There are many variants of the Piranha, and the in-service date is a guess.

Open APCs: The WRG 1950–1985 rules give class I (i.e., open) protection to the AT-P, BTR-50P, BTR-60P, and Half-tracks (M3).

8. **Armored Truck Platoon**

Mobility:

- The BTR-40 is based on the GAZ-63 truck.
- The BTR-152 is based on the ZIS-151 or ZIL-157 truck.
- The SPK/VPK are based on the Saab/Volvo flat-bed trucks.
- The Saxon has only four wheels, which makes me suspect it has limited mobility.

Light Armored Trucks: The WRG 1950–1985 rules give class II or III (i.e., light) protection to the BTR-152K. Beyond this, we have:

- Saxon: Wikipedia states that it is protected against small-arms rounds.

Open APCs: The WRG 1950–1985 rules give class I (i.e., open) protection to the BTR-40 and BTR-152. Beyond these, we have:

- SKP/VKP/SKPF/VKPF: Wikipedia states the armor is 8 to 20 mm, which is light.

9. **Armored Anti-Tank Missile Vehicles**

Anti-tank units are mainly chrome. However, in ground combat, we might give them more effectiveness against vehicular or towed targets and less against infantry targets.

Medium Anti-Tank Missile Vehicles: The WRG 1950–1985 rules give class IV (i.e., medium) protection to the Raketen JPz. Beyond this, we have:

- Jaguar: This is Raketen JPz (medium) upgraded with HOT missiles.

Light Anti-Tank Missile Vehicles: The WRG 1950–1985 rules give class II or III (i.e., light) protection to the BRDM and Striker. Beyond these, we have:

- Fennek MRAT: Based on the Fennek (light).
- FV438: This is derived from the FV432 APC (light).
- M150 and M901: These are derived from the M113 APC (light).
- M1134 Stryker ATGM: These are derived from the Stryker APC (light).
- VAB HOT: This is derived from the VAB APC (light).
- YP-408 PRAT: This is derived from the YPR-408 APC (light).
- YPR-765 PRAT: This is derived from the YPR-765 APC (light).

10. **Armored Reconnaissance Vehicles**

Again, reconnaissance vehicles are mainly chrome. However, in ground combat we might give them reduced effectiveness.

Light Reconnaissance Vehicles: The WRG 1950–1985 rules give class II or III (i.e., light) protection to the BRDM-2, Ferret, Lynx (M113½ Lynx), and M114. Beyond these, we have:

- LAV II Coyote: Wikipedia states that it is protected against small-arms rounds, which corresponds to light.
- Fennek: Wikipedia states that it is protected against small-arms rounds, which corresponds to light.

Light Reconnaissance Vehicles: The WRG 1950–1985 rules do not give any reconnaissance vehicles with class I (i.e., open) protection. However, we have:

- M20: Open version of the M8 (light) with no turret and armed only with a .50-cal machine gun.

11. **Armored Mortars**

For the protection class, we have:

- FV432: based on FV432 (light).
- M106/M125/M1064: based on M113 (light).
- M1129 Stryker MCV: based on Stryker (light).

We might consider these as open.

12. **Armored Artillery**

Light Armored Artillery: The WRG 1950–1985 rules give class II (i.e., light) protection to the Abbot, AMC-155GCT (AMX-30 AuF1), L33, M52, M108, M109, SP.73 (2S3), SP.74 (2S1), and VK.155 (BKan 1). Beyond these, we have:

- 2S19 Msta-S: Wikipedia states this all-round protection from 15 mm of steel armor. This corresponds to light armor.
- AS-90: Wikipedia states this has up to 17 mm of steel armor. This corresponds to light armor.
- Panzerhaubitze 2000: Wikipedia states this all-round protection from 14.5 mm of steel armor. This corresponds to light armor.

Open Armored Artillery: The WRG 1950–1985 rules give class I (i.e., open) protection to the AMX-105A, M7, and M44. Beyond these, we have:

- The M-50 has frontal and side armor, like the M7 and unlike for example the M107 and M108.
- The M40 also has frontal and side armor. The rear door is probably armored too, but is open when the gun is used.

13. **Armored Rocket Artillery**

Light Armored Rocket Artillery: For the protection class, we have:

- M270: I can find no definite information on the armor protection.
- BM-1: According to Wikipedia, the T-72 hull was chosen to give adequate protection close to the front lines. However, it is not clear whether the rocket launcher has adequate protection.

14. **Tracked Artillery**

To count as a hard target, both the crew and weapons must be protected by armor.

- The WRG 1950–1985 rules give class I (i.e., open) protection to the AMX.155F3 (AMX-13 F3), M107, and M110. However, the only crew member protected seems to be the driver. The other crew members travel and operate the gun without protection.
- Similar considerations apply to the 2S4, 2S7 (although some of the crew travel under armor), M40, M41, and Mk F3. In all of these, the gun mount is open.

15. **Mobile Artillery**

To count as a hard target, both the crew and weapons must be protected by armor.

- CAESAR: Some models have lightly-armored cabins, but the truck as a whole does not seem to be protected.
- Archer: The crew and engine compartments is protected against small-arms rounds, but the gun does not seem to be protected. Unlike the CAESAR, the gun is operated by the crew from the protected cabin. This is close to reaching the level of protection of light armored artillery.

16. **Mobile Rocket Artillery**

To count as a hard target, both the crew and weapons must be protected by armor.

- M142 HIMARS: The cabin has some protection, but the rest of the truck seems to be unprotected.
- BM-30: The cabin appears to have some protection, but the rest of the truck seems to be unprotected.
- Earlier Soviet vehicles appear to be completely unarmored.

17. **Mobile Rocket Artillery**

To count as a hard target, both the crew and weapons must be protected by armor.

- M752 Lance: This seems to be based on the unarmored M548.
- AMX-30 Pluton: The vehicle is based on an AMX-30 hull and seems to be armored, but the missile is exposed.

Tracked missile artillery is mainly included to serve as a target.

18. **Mobile Missile Artillery Batteries**

Missile artillery is mainly included to serve as a target.

Regarding the protection class, none of these have complete protection against small arms.

19. **AAA Values**

This long note summarizes how I have determined the AAA values for AAA units. VP values are discussed separately.

I compare AAA values for AAA units from *Air Strike*, *Eagles of the Gulf*, and *The Speed of Heat* to the actual characteristics of the guns they represent. My aim is to empirically understand the underlying model sufficiently to identify anomalous values and to apply it to other guns. That said, I do comment on the model.

AAA Values for Batteries: I start by attempting to understand the AAA values for batteries.

The upper part of the following table shows the original data for gun batteries from *Air Strike* (AS), *Eagles of the Gulf* (EOTG), and *The Speed of Heat* (TSOH). I have added columns with the caliber, burst rate of fire per mount, and muzzle velocity.

Type	Maximum Altitude	Ranges	Hit Rolls	Damage	FCR	Caliber (mm)	Rate of fire (RPM)	Muzzle Velocity (m/s)	Source
				Original					
M2/DShK-38	5	2/3/4	3/2/1	1	—	12.7	500	890	AS
M55	5	2/3/4	5/4/3	2	—	12.7	2000	890	EOTG
ZPU-1	6	2/3/5	3/2/1	1	—	14.5	600	1000	EOTG/TSOH
ZPU-2	6	2/3/5	4/3/2	1	—	14.5	1200	1000	AS
ZPU-4	6	2/3/5	5/4/3	2	—	14.5	2400	1000	AS
ZPU-4	6	2/3/5	4/3/2	2	—	14.5	2400	1000	TSOH
TCM-20	8	2/4/6	4/4/3	2	—	20	1450	860	EOTG
Single Rh-202	8	2/4/6	3/2/1	2	—	20	1030	1044	AS
Twin Rh-202	8	2/4/6	5/4/3	2	—	20	2060	1044	AS
M167	8	2/4/6	6/5/4	3	R	20	3000	1030	AS
ZU-23	9	2/5/8	4/3/2	2	—	23	2000	980	AS/TSOH
Gemag	10	3/6/9	6/5/4	4	R	25	1800	1040	AS
Oerlikon GDF	12	4/8/10	5/4/3	4	—	35	1100	1175	AS
61-K (M-38)	12	4/8/11	3/2/1	4	—	37	165	880	AS/TSOH
Bofors L/60	15	4/8/12	2/2/1	4	—	40	135	865	EOTG
Bofors L/70	15	4/8/12	3/3/2	4	—	40	330	1000	AS
BOFI	15	4/8/12	4/4/3	4	W	40	330	1000	AS
S-60	18	5/10/15	2/2/1	5	—	57	110	1000	AS/TSOH
KS-12	27	6/12/18	2/1/0	6	—	85	10	790	AS/TSOH
KS-19	39	7/14/21	1/1/0	6	—	100	15	900	AS
				Modified					
ZPU-4			5/4/3		— for rate of fire				
Single Rh-202			4/3/2		— for rate of fire				
Twin Rh-202				3	— for rate of fire				
ZU-23			5/4/3	3	— for rate of fire				
Oerlikon GDF			4/3/2		— for rate of fire				
BOFI			5/5/3		— for FCR-W				
				New					
M51	30	6/12/18	4/4/3	6	W	75	45	854	
90 mm M2	32	7/14/21	3/2/1	6	—	90	24	820	
QF 3.7 inch Mk VI	45	9/18/27	3/2/1	6	–	94	20	1044	
120 mm M1	57	11/22/33	2/2/1	7	–	120	12	945	
KS-30	64	12/24/36	1/1/0	7	—	130	12	970	

Qualitatively, altitude and range increase with caliber, hit rolls increase with rate of fire, and damage rating increases with both caliber and rate of fire. This is as expected.

Quantitatively:

- Approximately doubling the rate of fire increases the hit rolls by one, with typical values of 3/2/1 around 500 rounds per minute, 4/3/2 around 1000, and 5/3/2 around 2000 and above.
- The base damage rating is 1 for 12.7–14.5 mm, 2 for 20–23 mm, 3 for 25–30 mm, 4 for 35–40 mm, 5 for 57 mm, and 6 for 85–100 mm.
- The damage rating is increased by one if the rate of fire is 1800 rounds per minute or more.

Most of the original values can be used as they are, but a few of the hit rolls need adjusting:

- ZPU-4 and ZU-23
  There are six AAA batteries with rates of fire per mount of about 2000 rounds per minute or more: the M55, ZPU-4, twin Rh-202, M167, ZU-23, and Gemag. After accounting for the +1 modifier for the FCR-R of the M167 and Gemag, all have hit rolls of 5/4/3 except for the ZPU-4 from *The Speed of Heat* (although the one from *Air Strike* does have 5/4/3) and the ZU-23.
  Furthermore, all of these except for the ZU-23 (both *Air Strike* and *The Speed of Heat* versions) have damage ratings increased by one above their baseline values.
  It seems that either there is some reason to reduce the effectiveness of the ZPU-4 and ZU-23 compared to these other guns, or perhaps they have inconsistent values rolls.
  The ZU-23 is undeniably basic, and one might think that would be reason to reduce its effectiveness. However, the TCM-20 is basic too and has both a lower rate of fire than the ZU-23 (1450 compared to 2000) and a smaller caliber (20 mm compared to 23 mm), yet its hit rolls are higher than the ZU-23 in *The Speed of Heat* and its damage rating is the same.
  The best solution would seem to be to give these guns the values we would expect from comparing them to other guns with similar rate of fire and caliber. Therefore, I give both the ZPU-4 and ZU-23 hit rolls of 5/4/3 and the ZU-23 a damage rating of 3.
  (I also considered the idea that increasing the rate of fire increases *either* the hit rolls *or* the damage rating. However, while this might explain the ZPU-4 from *The Speed of Heat* and the twin Rh-202, it does not correctly predict the values of the M167 or Gemag.)
- Single RH-202
  The single Rh-202 has unexpectedly low original hit rolls of 3/2/1 for its rate of fire. The values of 4/3/2 for the ZPU-2 and TCM-20, which have a similar rate of fire, are probably more appropriate.
- Twin Rh-202
  The twin has an unexpectedly low original damage rating of 2 for its rate of fire. A value of 3, increased by one from the baseline for the caliber, is probably more appropriate.
- Oerlikon GDF
  The Oerlikon GDF has unexpectedly high original hit rolls of 5/4/3 for its rate of fire. The values of 4/3/2 for the ZPU-2, which has a similar rate of fire, are probably more appropriate.
- BOFI
  The BOFI has unexpectedly low original hit rolls of 4/4/3. Values of 5/5/3, obtained by applying a +2 modifier for FCR-W to the Bofors L/70, are probably more appropriate. (Note that the "BOFI" in *Air Strike* corresponds to what I call the "BOFI-R" below.)

I adopt these modified values.

Also, I note the following:

- FCR-R
  The M167 and Gemag have hit rolls of 6/5/4, and the twin Rh-202 has 5/4/3. Otherwise, they are similar, except that the M167 and Gemag have FCR-R, and the Rh-202 does not (and the M167 has a moderate advantage in rate of fire). I consider the FCR-R of the M167 and Gemag to be largely responsible for increasing their hit rolls by +1 compared to the Rh-202.
- Bofors
  The hit rolls of 2/2/1 for the Bofors L/60, 3/2/1 for the 61-K (M-38), and 3/3/2 for the Bofors L/70 seem to reflect the increase in rate of fire.
  The L/60 and L/70 have the same maximum altitude of 16 and maximum range of 12, despite the L/60 having lower muzzle velocity. However, I am not especially motivated to attempt to change this, since the Bofors is primarily a weapon for point-defense against aircraft up to about level 10.
  The real BOFI and BOFI-R also have proximity fuzes, and I consider these will increase their hit rolls by +1.

AAA Values for Sections: I now turn to sections. Again, my aim is to empirically understand the underlying model sufficiently to identify anomalous values and to apply it to other guns.

The upper part of the following table shows the original data for gun sections from *Air Strike* (AS), *Eagles of the Gulf* (EOTG), and *The Speed of Heat* (TSOH). Again, I have added columns with the caliber, burst rate of fire per mount, and muzzle velocity.

Type	Maximum Altitude	Ranges	Hit Rolls	Damage	FCR	Caliber (mm)	Rate of fire (RPM)	Muzzle Velocity (m/s)	Source
			Original						
M16	5	2/3/4	4/3/2	2	—	12.7	2000	890	EOTG
Mobile ZPU-4	6	2/3/5	3/3/2	2	—	14.5	2400	1000	AS
M163	8	2/4/6	5/4/3	3	R	20	3000	1030	AS
M3 TCM-20	8	2/4/6	3/3/2	2	—	20	1450	860	EOTG
Panhard	8	2/4/6	2/2/1	2	—	20	1450	860	AS
Mobile ZU-23	9	2/5/8	2/2/1	2	—	23	2000	980	AS
Nile 23	9	2/5/8	3/3/2	2	—	23	2000	980	EOTG
ZSU-23-4	9	2/5/8	5/4/3	3	W	23	4000	980	AS
Blazer	10	3/6/9	5/4/3	4	W	25	1800	1040	AS
M53/59	11	4/7/9	4/3/2	3	—	30	1000	1000	AS
AMX-30	11	4/7/9	5/4/3	3	W	30	1200	970	AS
ZSU-30-2	11	4/7/9	5/4/3	4	W	30	5000	960	AS
Gepard	12	4/8/10	5/4/3	4	W	35	1100	1175	AS
ZSU-57-2	18	5/10/15	2/1/1	5	—	57	240	1000	EOTG
			Modified						
Mobile ZPU-4			4/3/2		— to match the ZPU-4				
Panhard			3/3/2		— to match the M3 TCM-20				
Mobile ZU-23			4/3/2	3	— to match the ZU-23				
Nile 23			4/3/2	3	— to match the ZU-23				
M53/59			3/2/1		— to match the AMX-30				
			New						
AMX-13DCA	11	4/7/9	5/4/3	3	W	30	1200	970	
M19	15	4/8/12	2/2/1	4	—	40	270	865	
M42	15	4/8/12	2/2/1	4	—	40	270	865	

Comparing similar weapons, batteries and sections have the same maximum altitudes, ranges, and damage ratings. However, batteries (with four to six mounts) typically have a +1 advantage on their hit rolls compared to sections (with two mounts). In a very few cases, the difference is 0, and in a very few others, it is +2. A difference of 1 in the hit rolls is commensurate with the approximate doubling of the rate of fire from a section to a battery.

To determine the appropriate AAA for sections, the following recipe seems to work in most cases:

- Take the value for an equivalent battery without FCR and apply an adjustment of −1 to the hit rolls to account for the reduced rate of fire in the section. (Only do this if the rate of fire is reduced; see the comment on the ZSU-23-4 and ZSU-57-2 below.)
- Apply an adjustment of +1 to the hit rolls for FCR-R or similar.
- Apply an adjustment of +2 to the hit rolls for FCR-W.
- The preceding steps would then predict hit rolls of 6/5/4 for the Blazer and 7/6/5 for the ZSU-23-4 and ZSU-30-2. To conform to the original values, I limit the hit rolls to 5/4/3. It would appear that in the underlying model, beyond a certain rate of fire, other factors take over to limit the probability of a hit.

Again, most of the original data can be used as they are, but a few of the values are worth commenting upon or need adjusting.

- Agreements
  The original hit rolls and the predicted hit rolls are in agreement for the M16 (from the M55 battery), M3 TCM-20 (from the TCM-20 battery), M163 (from the M167 battery), the Blazer (from the Gemag), and the Gepard (from the Oerlikon GDF).

- Mobile ZPU-4
  The original hit rolls for the mobile ZPU-4 section are 3/3/2, but 4/3/2 is predicted from the values of 5/4/3 for the battery and is in better agreement with the values for the M16.
- Panhard M3 VDA
  The Panhard M3 VDA has unexpectedly low original hit rolls of 2/2/1 compared to the TCM-20 battery and the M3 TCM-20 section, despite having very similar guns. Values of 3/3/2 are probably more appropriate.
- Mobile ZU-23 and Nile 23
  The original hit rolls for the mobile ZU-23 and Nile 23 section are 2/2/1 and 3/3/2, despite them being quite similar. The predicted values of 4/3/2 are probably more appropriate and are in better agreement with the values for the M16. Also, as with the ZU-23 battery, a damage rating of 3 is probably more appropriate given their high rate of fire.
- M53/59
  The original hit rolls for the M53/59 section are 4/3/2. The predicted values of 3/2/1, derived from the modified hit rolls of 4/3/2 for the Oerlikon GDF battery and also from the AMX-30SA after discounting the FCR-W, are probably more appropriate.
- ZSU-57-2
  The ZSU-57-2 section has the same hit rolls as the S-60 battery. This is probably appropriate, as the ZSU-57-2 is a twin mount and this largely compensates for the reduced rate of fire of the section. The same applies to the quad ZSU-23-4 compared to the twin ZU-23.

I adopt these modified values.

Also, I note:

- Panhard M3 VDA
  The Panhard M3 VDA has an FCR-R capability, which is discussed below but not considered here.
- AMX-30 DCA
  The real AMX-30 DCA does not have an FCR. The AMX-30 DCA in the game seems to correspond to the similar AMX-30SA, which does.

The AAA values for new guns are discussed when they are present.

Summary: I have developed an empirical understanding of the model for AAA values. Consideration of outliers suggests:

- Increasing in the hit rolls of the mounted ZPU-4, single Rh-202, Panhard M3 VDA, ZU-23 (and derivatives), and BOFI.
- Decreasing the hit rolls of the Oerlikon GDF.
- Increasing the damage rating of the twin Rh-202 and the ZU-23 guns (and derivatives).

I have adopted these changes.

20. **M2 Platoon**

The AAA values are from *Air Strike*.

Sources:

- Wikipedia on the M2.
- WeaponSystems on the M2.

21. **DShK-38 Platoon**

The AAA values are from *Air Strike*.

Service in the Korean War is from Singh, ch. 1.

Sources:

- Wikipedia on the DShK.
- WeaponSystems on the DShK.
- Singh, M. (Pen & Sword, 2020): "Anti-Aircraft Artillery in Combat 1950–1972: Air Defence in the Jet Age".

22. **M16 Section**

The AAA values are from *Eagles of the Gulf*.

Sources:

- Wikipedia on the M16, M45, and M2.
- WeaponSystems on the M16 and M2.

23. **M16 Section**

The AAA values are the same as for the M16.

Sources:

- Wikipedia on the M17, M45, and M2.
- Tank Encylopedia on the M17 .

24. **M55 Battery**

The AAA values are from *Eagles of the Gulf*.

Sources:

- Wikipedia on the M45 and M2.
- WeaponSystems on the M2.

25. **Vz.53 Platoon**

The AAA values the same as for the M55.

Sources:

- Wikipedia on the Vs.53 and DShK.
- WeapoNEWS on the Vz.53.
- WeaponSystems on the DShK.

26. **BTR-152A Vz.53 Section**

I have given this the armor and mobility of the M16.

The in-service date assumes they were supplied or converted shortly after the 1955 arms deal. Some *Eagles of the Gulf* scenarios for 1956 War give the Egyptian Army M16s, but I do not think these served with the Egyptian Army; I think these might well be BTR.152A Vz.53s.

Sources:

- Wikipedia on the BTR-152A, Vs.53, and DShK.
- WeapoNEWS on the Vz.53.
- Army Guide on the BTR-152.
- Miranda, M., 2015: "Gun Truck"

27. **M55 Battery**

The AAA values are from *Eagles of the Gulf*.

Sources:

- Wikipedia on the M45, M2, and gun trucks.
- WeaponSystems on the M2.

28. **ZPU-1/2/4 Battery**

The AAA values for the ZPU-1 are from *Eagles of the Gulf* and *The Speed of Heat*. The AAA values for the ZPU-2 are from *Air Strike*. The AAA values for the ZPU-4 are from *Air Strike* rather than *The Speed of Heat*, for the reasons described in the extended note above.

Sources:

- Wikipedia on the ZPU.
- WeaponSystems on the ZPU-1, ZPU-2, and ZPU-4, and KPV.

29. **Mobile ZPU-1/2/4 Section**

AAA values for the mobile ZPU-4 are from *Air Strike*, except that I have adopted hit rolls of 4/3/2 rather than 3/3/2 as described in the extended note above. The mobile ZPU-1/2 do not appear in *Air Strike*, *Eagles of the Gulf*, or *The Speed of Heat*, but they are very common. As described in the extended note above, I have produced AAA hit values for the sections from those of the ZPU-1 and ZPU-2 batteries.

Sources:

- Wikipedia on the ZPU and Technicals.
- WeaponSystems on the ZPU-1, ZPU-2, and ZPU-4, and KPV.

30. **BTR-40A Section**

I have given this the armor and mobility of the M16 and the AAA properties of the mobile ZPU-2 section.

The BTR-40 is based on a truck and does not seem to have especially good mobility; we might consider giving it "G/P" mobility rather than "U".

Sources:

- Wikipedia on the BTR-40A and ZPU-2.

31. **BTR-152A Section**

I have given this the armor and mobility of the M16 and the AAA properties of the mobile ZPU-2/4 section.

The BTR-152 is based on a truck and does not seem to have especially good mobility; we might consider giving it "G/P" mobility rather than "U".

Army Guide states that the A/E mounted a ZPU-2 and the D a ZPU-4. Others (Wikipedia) that the A/D/E referred to the chassis (with the D using the BTR-152V and the E the BTR-152V1 chassis) rather than the gun mount. I am not sure how to resolve this, but I don't think it is that important. Instead, I will use "BTR-152A ZPU-2" and "BTR-152A ZPU-4," which extends obviously to other guns mounted on the BTR-152A and similar.

Sources:

– Wikipedia on the BTR-152A and ZPU-2.
– Army Guide on the BTR-152.
– Miranda, M., 2015: "Gun Truck"

32. **M167 Platoon**

The AAA values are from *Air Strike*.

The 1976 Operator's Manual describes the use of an AN/TVS-2B or AN/TVS-5 night sight. These are first- and second-generation image intensifiers. They are mounted next to the tracking sight and appear to be useful only for ground targets.

Sources:

– Wikipedia on the M167 and M61 Vulcan.
– WeaponSystems on the M167 and M61.
– Department of the Army, TM 9-1005-286-10, 1976: "Operator's Manual (Crew) for Gun, Air Defense Artillery, Towed".

33. **M163 Section**

The AAA values are from *Air Strike*.

The 1976 Operator's Manual describes the use of an AN/VPS-2 range-only radar.

The 1976 Operator's Manual describes the use of an AN/TVS-2B night sight. This is a first-generation image intensifier. It is mounted next to the tracking sight and appears to be useful only for ground targets.

The PIVADS version introduced in 1984 had an improved sight, but

Sources:

– Wikipedia on the M163, M167, and M61 Vulcan.
– WeaponSystems on the M163, M167 and M61.
– Department of the Army, TM 9-2350-300-10, 1976: "Operator's Manual For Gun, Air Defense Artillery, Self-Propelled 20-mm, M163".

34. **TCM-20 Platoon**

The AAA values are from *Eagles of the Gulf*.

Sources:

– Wikipedia on the TCM-20 and HS.404.
– WeaponSystems on the TCM-20.

35. **M3 TCM-20 Section**

The AAA values are from *Eagles of the Gulf*.

Sources:

– Wikipedia on the M3 TCM-20, TCM-20, HS.404.
– WeaponSystems on the M3 TCM-20 and TCM-20.

36. **Rh-202 Battery**

The AAA values are from *Air Strike*, with the hit values for the single mount changed from 3/2/1 to 4/3/2 and the damage value for the twin mount changed from 2 to 3, as described in the extended note above.

Air Strike has single and twin Rh-202 batteries. Single mounts seem to be used for AFVs (e.g., Marder, Luchs, and Wiesel) and naval applications, but all AAA mounts seem to be twin.

Sources:

– Wikipedia on Rh-202.
– WeaponSystems on the Rh-202 twin mount and Rh-202 gun.

37. **Panhard M3 VDA Section**

The Panhard M3 VDA in *Air Strike* seems to correspond to Wikipedia M3 VDA or WeaponSystems M3 VDAA.

The AAA data are from *Air Strike*, but modified as described in the extended note above to give hit rolls of 3/3/2 without FCR. However, the Panhard M3 VDA radar-equipped vehicle seems to be able to search and share speed and range information with the other vehicle in the section, which seems equivalent to FCR-R. Therefore, I have increased the hit rolls by one to give 4/4/3 with FCR-R.

Sources:

– Wikipedia on the Panhard M3 VDA and HS.820.
– WeaponSystems on the Panhard M3 VDAA.

38. **ZU-23 Battery**

The AAA values are from *Air Strike*, but the hit rolls are modified as from 4/3/2 to 5/3/2 and the damage rating from 2 to 3 as described in the extended note above.

Sources:

– Wikipedia on the ZU-23.
– WeaponSystems on the ZU-23 and 2A14.

39. **Mobile ZU-23 Section**

The AAA values are from *Air Strike* and *The Speed of Heat*, but the hit rolls of 4/3/2 were obtained from those for the ZU-23 battery using the recipe described above, and the damage was modified from 2 to 3 as described in the extended note above.

Sources:

– Wikipedia on the ZU-23 and Technicals.
– WeaponSystems on the ZU-23 and 2A14.

40. **BTR-152A ZU-23 Section**

I have given this the armor and mobility of the M16 and the AAA properties of the mobile ZPU-2/4 section.

The BTR-152 is based on a truck and does not seem to have especially good mobility; we might consider giving it "G/P" mobility rather than "U".

Army Guide states that the A/E mounted a ZPU-2 and the D a ZPU-4. Others (Wikipedia) that the A/D/E referred to the chassis (with the D using the BTR-152V and the E the BTR-152V1 chassis) rather than the gun mount. I am not sure how to resolve this, but I don't think it is that important. Instead, I will use "BTR-152A ZPU-2" and "BTR-152A ZPU-4," which extends obviously to other guns mounted on the BTR-152A and similar.

Sources:

– Wikipedia on the BTR-152A and ZU-23.
– Army Guide on the BTR-152.
– Miranda, M., 2015: "Gun Truck"

41. **ZSU-23-4 Section**

The AAA values are from *Air Strike*.

Sources:

– Wikipedia on the ZSU-23-4.
– WeaponSystems on the ZSU-23-4 and 2A7.

42. **Sinai 23 Section**

The gun properties are the same as the mobile ZU-23, but with +1 to the hit rolls for the FCR-R. This means that the Sinai 23 (with two barrels) then has the same hit rolls and damage rating as the ZSU-23-4 (with four barrels). Perhaps this is acceptable, since there are 30 years between these systems.

The turret is the Dassault TA-23E and seems to be similar to the LAV-AD turret, but with a pair of 23 mm guns instead of a GAU-12 25 mm gun. The AAA values were taken from the mobile ZU-23 battery. I don't know which model of Stinger was used; I would guess the FIM-92A.

The brochure explicitly mentions a "day sight"; there is no sign of an IR sight.

A platoon seems to consist of three "satellite" fire units and one "leader" unit with a RA 20S radar. The radar has only 1 degree precision, so it is not a traditional FCR, but seems to give warning, cueing, and range for optical tracking. I will treat it as an FCR-R and give a +1 modifier.

The radar works in the E-band. This is 2-3 GHz or UHF.

Perhaps the most accurate way to handle the radar would be to have a platoon of 3 fire units and 1 radar unit, and for each D determine if the radar unit is killed on a 3–.

Sources:

– Wikipedia on Sinai 23 and ZU-23.
– Brochure describing the Sinai 23.
– Description of RA 20S FCR.

43. **LAV-AD Section**

The system seems to be similar to the Blazer, but without radar and with a laser range finder and night IR sights. I've adopted the range and damage rating for the Gemag battery in *Air Strike* with the hit rolls reduced by one for being a section. The laser rangefinder substitutes for the FCR-R, so the hit rolls are not reduced further.

Sources:

– Wikipedia on the LAV-AD, GAU-12, and FIM-92.
– Dickerson, C.R., 1991: "LAV-AD: Mobile Air Defense for the Ground Maneuver Force".

44. **M53/59 Section**

The AAA values are from *Air Strike*, but the hit rolls were reduced from 4/3/2 to 3/2/1 as described in the extended note above. If the VADS is classed as armored, should this be armored too?

Sources:

– Wikipedia on the M53/59.
– WeaponSystems on the M53/59 and M53.

45. **AMX-30SA Section**

I adopt the *Air Strike* values given for the AMX-30 DCA.

Sources:

– Wikipedia on the AMX-13DCA and HS.831A.
– WeaponSystems on the AMX-13DCA HS.831A.

46. **AMX-30SA Section**

I adopt the *Air Strike* values given for the AMX-30 DCA. The real DCA does not have a tracking radar, but the *Air Strike* one does, matching the SA.

Sources:

– Wikipedia on the AMX-30DCA/SA and HS.831A.
– WeaponSystems on the AMX-30DCA HS.831A.

47. **Tunguska Section**

The AAA values are from *Air Strike* for the ZSU-30-2.

Sources:

– Wikipedia on the Tunguska.
– WeaponSystems on the Tunguska, 2A38, and 9M311.

48. **Tunguska-M Section**

The AAA values are from *Air Strike* for the ZSU-30-2.

Sources:

– Wikipedia on the Tunguska.
– WeaponSystems on the Tunguska, 2A38, and 9M311.

49. **Oerlikon GDF Battery**

The AAA values for the battery are from *Air Strike* with the hit rolls reduced from 5/4/3 to 4/3/2 as described in the extended note above.

Sources:

– Wikipedia on the GDF.
– WeaponSystems on the GDF and KDA.

50. **Gepard Section**

The AAA values are from *Air Strike*.

Sources:

– Wikipedia on the Gepard.
– WeaponSystems on the Gepard and KDA.

51. **61-K Battery**

The AAA values are from *Air Strike* and *The Speed of Heat*.

I can find no sources other than *Air Strike*, *Eagles of the Gulf*, and *The Speed of Heat* that refer to the 61-K 37 mm gun as the "M-38". Another designation for the gun is "M-1939", so I could understand "M-39".

Sources:

– Wikipedia on the 61-K.
– WeaponSystems on the 61-K.

52. **Bofors L/60 Battery**

The AAA values are from *Eagles of the Gulf*.

Sources:

– Wikipedia on the Bofors L/60.
– WeaponSystems on the Bofors L/60.

53. **Bofors L/60 Battery**

The AAA values are identical to those of the Bofors L/60 battery, since the number of guns is the same.

Sources:

– Wikipedia on the M19.

54. **Bofors L/70 Battery**

The AAA values are from *Air Strike*.

Singh (ch. 4) describes the Israeli use.

Sources:

– Wikipedia on the Bofors L/70.
– WeaponSystems on the Bofors L/70.
– Singh, M. (Pen & Sword, 2020): "Anti-Aircraft Artillery in Combat 1950–1972: Air Defence in the Jet Age".

55. **Bofors L/60 Battery**

The AAA values are identical to those of the Bofors L/60 battery, since the number of guns is the same.

Sources:

– Wikipedia on the M42.

56. **Bofors L/70 BOFI Battery**

The AAA values are adapted from *Air Strike* values for the Bofors L/70, with the hit rolls increased by one for proximity fuzes and one for the BOFI laser rangefinder (effectively FCR-R), giving hit rolls of 5/5/4 for BOFI.

I don't know when the BOFI became available or who else uses them.

Sources:

– Wikipedia on the BOFI and BOFI-R.
– WeaponSystems on the BOFI and BOFI-R.

57. **Bofors L/70 BOFI-R Battery**

The AAA values are adapted from *Air Strike* values for the Bofors L/70, with the hit rolls increased by one for proximity fuzes and another two for the BOFI-R radar, giving hit rolls of 6/6/5 for BOFI-R whereas *Air Strike* gives the BOFI-R 5/5/4.

I don't know when the BOFI-R became available or who else uses them.

Sources:

– Wikipedia on the BOFI and BOFI-R.
– WeaponSystems on the BOFI and BOFI-R.

58. **S-60 Battery**

The AAA values are from *Air Strike* and *The Speed of Heat*.

Sources:

– Wikipedia on the S-60.
– WeaponSystems on the S-60.

59. **ZSU-57-2 Section**

For the ZSU-57-2 section: The AAA values are from *Eagles of the Gulf*.

Sources:

– Wikipedia on the ZSU-57-2.
– WeaponSystems on the ZSU-57-2.

60. **M51 Battery**

Wikipedia gives a rate of fire of 45 rounds per minute, a muzzle velocity of 854 m/s, an effective maximum altitude of 30k feet, and a horizontal range of 26 hexes (13 km).

The TM states that the search radar is X-band (6–11 GHz or SHF).

The AAA hit rolls are based on the S-60, reduced by 1 for a lower rate of fire, increased by 1 for proximity fuzes, and increased by 2 for FCR-W, so 4/4/3. I will give it the same range of 6/12/18 as the 85 mm JS-12, which fires a slightly heavier round at a slightly lower velocity.

Sources:

– Wikipedia on the M51 and T38 radar
– WeaponSystems on the M51.
– Werrell, K.P. (2005, Air University Press, 2nd edition): "Archie to SAM".
– Departament of the Army (1956): TM 9-6092-1-1 "Antiaircraft Fire Control System M33C and M33D Operation, June 1956".

61. **KS-12 Battery**

The AAA values are from *Air Strike* and *The Speed of Heat*.

There are two KS-12 guns. The M-1939 is given a maximum altitude of 34k feet by Wikipedia and 25k feet by Singh. The M-1944 is given a maximum altitude of 38k ft by Wikipedia 29k ft by Singh. The game version have maximum altitudes of 27k feet, which is half-way between the figures given by Singh.

Wikipedia gives maximum horizontal ranges of 15.65 km and 18 km for the M-1939 and M-1944.

Wikipedia gives the rate of fire as 10–12 RPM.

It is not clear to me that the Soviets had proximity fuzes for this gun.

Sources:

– Wikipedia on the KS-12.
– WeaponSystems on the KS-12.
– Singh, M. (Pen & Sword, 2020): "Anti-Aircraft Artillery in Combat 1950–1972: Air Defence in the Jet Age".

62. **90 mm M2**

The M1 version entered service in 1940. The M2 was a development of the M1 and was the "standard weapon" from 1943. I presume all post-war examples are M2s.

Wikipedia gives the rate of fire as 24 RPM, about twice that of the KS-12.

Wikipedia gives a horizontal range of 19 km, and vertical range of 43.5k feet. Warrell states the effective ceiling is 32k feet. As the calibre and the muzzle velocity of the M2 are between the KS-12 (maximum altitude of 27) and the KS-19 (maximum altitude of 39), this seems to be about right.

I give the M2 the hit rolls of the KS-12 increased by 1, to account for the increased rate of fire and proximity fuzes (although the number of guns per battery is reduced from six to four), a maximum altitude of 32 (following Warrell), and the ranges of the KS-19.

Sources:

– Wikipedia on the 90 mm M2.
– Werrell, K.P. (2005, Air University Press, 2nd edition): "Archie to SAM".

63. **QF 3.7 inch Battery**

Wikipedia gives a maximum altitude of 45k feet, a muzzle velocity of 1044 m/s (significantly more than other similar guns), and a rate of fire of 20 rounds per minute.

I give the QF 3.7 the hit rolls of the KS-12 increased by 1, to account for the increased rate of fire and proximity fuzes, a maximum altitude of 45, and a maximum range of 27.

Singh states that by 1956 the IDF had twelve 3.7 inch guns with FCR.

Singh states that Pakistan used the 3.7 inch gun in the 1971 War.

Sources:

– Wikipedia on the QF 3.7 inch.
– Singh, M. (Pen & Sword, 2020): "Anti-Aircraft Artillery in Combat 1950–1972: Air Defence in the Jet Age".

64. **KS-19 Battery**

The AAA values are from *Air Strike*.

Wikipedia gives a rate of fire of 15 RPM, a vertical range of 48 altitude levels (15 km, with a proximity fuse), and a horizontal range of 39 hexes (21 km). The *Air Strike* values are 39 altitude levels and 21 hexes, so are cut back somewhat from the maximums.

It is not clear to me that the Soviets had proximity fuzes for this gun. The one suggestion that they do is that Wikipedia mentions different maximum altitudes for proximity and timed fuzes, but I can find no information on Soviet proximity fuzes.

Sources:

– Wikipedia on the KS-19.
– WeaponSystems on the KS-19.

65. **120 mm M1 Battery**

Wikipedia gives rate of fire of 12 RPM, a maximum altitude of 57 altitude levels (17.5 km), and a maximum range of 50 hexes (25 km).

The hit rolls are adapted from the KS-30 (which has a similar rate of fire), with a +1 modifier for proximity shells (although the number of guns per battery is reduced from six to four).

Sources:

– Wikipedia on the M1.

66. **KS-30 Battery**

Wikipedia gives the rate of fire is 10–12 RPM, the vertical range as 72 altitude levels (22 km) and 54 hexes (29 km) horizontally. Wikipedia also says it provided a 25% increase over the effective ceiling of the KS-19, which would be 48 altitude levels. However, this has a heavier shell and slightly higher muzzle velocity than the 120 mm M1, so it should have a maximum altitude of more than 57. I will give it a maximum altitude of 64.

It is not clear to me that the Soviets had proximity fuzes for this gun.

The hit rolls are adapted from the KS-19 (which has a similar rate of fire). In order to reach an altitude of 64, it needs a maximum range of at least 32. I will give it 36.

Sources:

– Wikipedia on the KS-30.